

Review

Advanced Chip-Level Thermal Management Technologies for High-Power Integrated Processors: A Review

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Abstract

The power density of modern high-power integrated processors keeps rising rapidly. Among them, chiplet-based high-power AI accelerators exhibit local peak heat flux exceeding 1 kW/cm^2 , which leads to concentrated hotspots, severe internal temperature gradients, device performance degradation and reliability deterioration. Conventional heat dissipation approaches are limited by the bottleneck of series interfacial thermal resistance and fail to meet the cooling demands of complex integrated architectures. Chip-level thermal management serves as a core method to suppress hotspots near heat sources and reduce overall system thermal resistance, which guarantees long-term stable operation of high-power integrated processors and plays a vital role in improving the energy efficiency and service life of computing platforms. This paper systematically reviews mainstream chip-level thermal management technologies for high-power integrated processors, covering heterogeneous integration of high-thermal-conductivity substrates, embedded microchannel liquid cooling, solid-state active heat pumps, multi-physics co-design and advanced packaging manufacturing processes. The basic working principles and state-of-the-art research progress of each cooling technology are elaborated in detail. The common engineering bottlenecks, including ultra-high heat flux endurance, packaging process compatibility, fluid leakage risks and multi-layer interfacial thermal resistance, are summarized, and the future development trends of this field are clarified. This review can provide comprehensive theoretical guidance for structural design and large-scale engineering implementation of near-junction thermal management solutions for various high-power integrated processors, especially high-computing-power AI accelerators.

Keywords: high-power integrated processors; high-power AI chips; chip-level thermal management; high-thermal-conductivity materials; microchannel cooling; solid-state refrigeration; multi-physics coupling



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1. Introduction

1.1. Research Background

With the rapid evolution of integrated circuit technology toward heterogeneous chiplet integration, high-power power electronics and high-computing large-model scenarios, the power density of various high-power integrated processors (HPIPs) rises exponentially.

Among them, high-power AI accelerators for large-model training act as typical representatives of HPIPs: the thermal design power (TDP) of a single mainstream training chip exceeds 700 W [1], and the local heat flux density in high-density computing regions can surpass 300 W/cm² [2], far exceeding the heat dissipation design threshold of conventional general-purpose processors. Meanwhile, wide-bandgap high-power integrated power devices such as silicon carbide (SiC) and gallium nitride (GaN) operate under high-voltage and high-heat-flux conditions, which jointly impose stringent requirements on existing cooling systems. The sharp rise in device power consumption mainly originates from the dense integration of tensor computing units driven by expanding parameters of large AI models, elevated power density of power electronic devices, and interlayer thermal coupling effects induced by through-silicon via (TSV) in 3D stacking and multi-chiplet heterogeneous packaging. The alternating operation of high bandwidth memory (HBM) and tensor cores further intensifies transient local hotspots, leading to highly spatiotemporally uneven heat generation across the entire chip under diverse operating conditions.

Excessive power consumption directly increases the core junction temperature of device active regions. High-temperature conditions aggravate transistor leakage current loss, shorten device service life and degrade long-term operational reliability. In addition, the built-in dynamic frequency scaling protection mechanism will be triggered, which directly limits the rated performance output of HPIPs. Furthermore, the thermal conductivity of traditional silicon substrates is merely around 150 W/(m·K), and the coefficient of thermal expansion (CTE) of silicon fails to match thin-film electrodes and interposer materials. A standalone silicon substrate cannot rapidly dissipate concentrated hotspot heat under integrated architectures; the absence of high-performance thermal interface material (TIM) will further enlarge interfacial thermal resistance, forming a universal technical bottleneck restricting continuous improvement of overall power density. Therefore, developing advanced chip-level thermal management technologies, breaking the inherent thermal conductivity limitations of silicon-based materials, and constructing efficient global heat conduction and exchange channels serve as the core foundation to guarantee long-term stable operation of all types of HPIPs and support performance iteration of both AI computing chips and wide-bandgap power devices. Such technologies are also critical for reducing the power usage effectiveness (PUE) and improving the comprehensive energy efficiency of data center systems.

1.2. Strategic Importance of Thermal Management for High-Power Integrated Processors

With the rapid development of high-end integrated circuits, thermal management for high-power integrated processors (HPIPs) has evolved from a conventional auxiliary technique into a core technology that determines chip performance, operational reliability, and industrial competitiveness. According to the U.S. Air Force Avionics Integrity Program (AVIP), temperature-related stress accounts for approximately 55% of electronic equipment failures. Therefore, thermal control remains a critical reliability requirement for HPIPs operating under high-power-density and highly integrated conditions [3]. High temperature accelerates electromigration, negative bias temperature instability, hot carrier injection, and other degradation mechanisms, causing transistor aging, parameter drift, and permanent performance deterioration.

As the most thermally stringent subset of HPIPs, high-power AI accelerators face severe thermal challenges. The thermal design power (TDP) of state-of-the-art AI accelerators has reached the kilowatt level. For instance, the NVIDIA Blackwell architecture achieves a chip GPU exceeding 1000 W [4], and the overall module thermal load approaches 1200 W when integrated with high bandwidth memory (HBM) and dense interconnection components [5]. The drastically increased on-chip thermal stress and temperature gradi-

ents aggravate interfacial reliability risks induced by mismatched coefficient of thermal expansion (CTE) among packaging materials.

Industrial thermal failure cases have fully demonstrated such reliability risks. The RTX 50 series has suffered from functional defects [6], poor contact of thermal interface materials (TIMs), and thermal gel overflow, resulting in local overheating in memory and power regions and adversely affecting product stability [7,8]. These incidents verify that thermal bottlenecks have become critical constraints on the mass production and performance release of modern HPIPs [6–8].

Currently, advanced chip thermal management has become a mainstream industrial focus. Leading manufacturers including TSMC, Samsung, and Intel regard embedded on-chip cooling, microchannel heat dissipation, and heterogeneous thermal co-design as key directions for next-generation high-performance chips. The global chip thermal management market is projected to grow from 19.8 billion USD in 2025 to 30 billion USD in 2030, with a compound annual growth rate of 8.6% [9]. It is concluded that refined chip-level thermal management serves as an essential foundation for breaking high heat flux limitations, improving energy efficiency and service life, and supporting the continuous iteration of high-power integrated processors.

1.3. Review Scope and Structure

This review targets advanced chip-level thermal management technologies for HPIPs. High-power AI accelerators are taken as the core research representative within the HPIP category, while wide-bandgap power electronic devices are also discussed as supplementary research objects. The overall research scope falls into three major branches: near-junction liquid cooling, packaging thermal management, and thermal performance optimization of functional materials. Detailed contents cover core technologies including microchannel liquid cooling, jet impingement cooling, thermal interface materials (TIMs), heat pipes and vapor chambers, high-conductivity substrate integration, as well as collaborative thermal design tailored for chiplet and three-dimensional integrated circuit (3D IC) architectures.

The literature search was conducted using the core keywords “High-power AI processors”, “Chip-level thermal management”, “Microchannel cooling”, “Thermal interface materials”, and “High thermal conductivity substrates”. The databases searched include Web of Science, IEEE Xplore, ScienceDirect, and CNKI (China National Knowledge Infrastructure). The selected literature covers SCI/EI journal articles, high-level conference papers, and key Chinese core journals from the past two decades, ensuring both technological currency and authority.

The full text follows a progressive logic from problem to solution to outlook: Section 1 provides an introduction, elaborating on the research background of thermal dissipation in high-power AI chips and the strategic importance of thermal management, while clarifying the scope of the review and the structural framework. Section 2 analyzes the thermal challenges of high-power AI chips, focusing on the increasing trends in power consumption and heat flux density, as well as the thermal difficulties introduced by chiplet/3D IC architectures, and demonstrates the limitations of traditional cooling solutions. Section 3 systematically classifies and reviews the principles, research status, and performance characteristics of core technologies in near-junction liquid cooling, package-level thermal management, and material optimization. Section 4 compares the advantages, disadvantages, and applicable scenarios of various technologies, and identifies key bottlenecks currently facing thermal management. Section 5 summarizes the research progress of the entire paper and provides an outlook on the future development directions of thermal management technologies for high-power AI chips.

1.4. Literature Review Methodology

To comprehensively sort out advanced chip-level thermal management technologies for high-power heterogeneous integrated chips, this review systematically retrieves the relevant literature from mainstream databases including Web of Science, IEEE Xplore, ScienceDirect and CNKI. The retrieval time range covers publications issued from 1981 to 2026. The core search keywords are defined as follows: high-power chip, heterogeneous integration, chiplet, near-junction cooling, thermal interface material, immersion cooling, microchannel heat sink, thermal characterization, wide-bandgap device, AI processor.

Clear inclusion and exclusion criteria were formulated for literature screening. The inclusion criteria are as follows: (1) peer-reviewed journal articles and conference papers focusing on chip-level thermal management, packaging heat dissipation and thermal measurement technology; (2) studies with quantitative experimental data, numerical simulation results or comparative performance analysis of cooling schemes; (3) research targeting high-power heterogeneous chips, AI accelerators and wide-bandgap power integrated devices. The exclusion criteria include the following: (1) patents, book chapters, review papers without original comparison data; (2) studies only focusing on system-level data center energy optimization without chip-scale heat dissipation analysis; (3) articles with repeated research content or insufficient technical details.

After preliminary keyword retrieval, a total of 2168 papers were initially obtained. Duplicates were removed first, followed by two rounds of screening by reading titles, abstracts and full texts sequentially. Finally, 179 valid studies were retained for in-depth sorting and discussion. According to the research focus, the selected literature was classified into four major categories: near-junction liquid cooling technologies, packaging thermal management materials and structures, thermal characterization and temperature measurement methods, and thermal optimization oriented to heterogeneous chip workloads. The whole literature screening process is visually presented in the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow diagram (Figure 1). To provide a structured overview of the representative studies identified through this screening process, the core research contributions of the selected literature are further summarized in Table 1, categorized by technology roadmap.

Table 1. Summary of references.

Technology Roadmap Category	Core Authors	Publication Year Range	Core Research Chip Type	Summary of Core Performance Metrics
Basic Theory of Chip Thermal Management and Industry Development Trend	NVIDIA, IEEE Electronics Packaging Society, ESMAEILZADEH H, et al.	2012–2023	H100 Tensor Core AI GPU, Heterogeneous Integration Chip/2.5D/3D Chiplet, Multi-Core CPU/GPU General-Purpose Processor.	Clarified the exponential growth trend of thermal load of high-power AI chips, and predicted that the local heat flux density of chips will exceed 1 kW/cm ² by 2030.
Microchannel Liquid Cooling Technology	TUCKERMAN D B, Tang W, KONG D, Jung K W, et al.	1981–2023	VLSI Very Large Scale Integration Circuit Chip, High-Power CPU/GPU Processor, High-Power AI Accelerator Chip, 3D Stacked IC Chip.	The thermal resistance of the foundational microchannel heat sink structure is as low as 0.1 °C/W; the heat transfer efficiency of the manifold microchannel is 40% higher than that of the traditional structure, and the flow resistance is reduced by 25%; the additively manufactured structure can stably carry a high heat flux density of 1 kW/cm ² ; the thermal resistance of the on-chip embedded structure is 30% lower than that of the traditional microchannel.

Table 1. Cont.

Technology Roadmap Category	Core Authors	Publication Year Range	Core Research Chip Type	Summary of Core Performance Metrics
Jet Impingement Cooling Technology	Stefanescu S, Wei T, et al.	1999–2022	Power Electronic Chip, High-Power AI Chip.	The heat transfer coefficient of the microjet array exceeds $10^5 \text{ W}/(\text{m}^2 \cdot \text{K})$; the full-chip temperature difference of the alternating inlet and outlet jet array is 50% lower than that of the traditional jet chip, and the crossflow interference is completely eliminated.
Cold Plate Liquid Cooling and Immersion Liquid Cooling Technology	Jarrett A, Bazmi B, Lu J, et al.	2011–2026	Power Device/Power Battery, High-Power AI Accelerator Card, Flagship AI Chip.	The thermal resistance of the optimized cold plate is reduced by 25%, and the temperature uniformity is improved by 40%; the topology-optimized additively manufactured cold plate has 30% lower thermal resistance and 68% lower pressure drop than the traditional cold plate; the single-chip heat dissipation capacity of passive two-phase immersion cooling exceeds 1000 W.
Thermal Interface Material (TIM) Technology	Wu K, Zhang Y, et al.	2023–2025	High-Power AI Chip, High-Power GaN/AI Chip.	The interfacial thermal resistance of liquid metal TIM is as low as $0.05 \text{ K} \cdot \text{cm}^2/\text{W}$; the intrinsic thermal conductivity of diamond composite TIM exceeds $1000 \text{ W}/(\text{m} \cdot \text{K})$, which greatly reduces the interfacial thermal resistance.
Package-Level Passive Thermal Management Technology	Grover G M, Jiang H, et al.	1964–2024	Semiconductor Device, High-Power GPU/NPU.	The thermal conductivity of the heat pipe is far higher than that of solid metal, enabling long-distance efficient heat transfer; the in-plane temperature difference of the ultra-thin vapor chamber is 60% lower than that of the traditional structure.
Advanced Packaging and Thermal Co-Design Technology	Singh S G, Shi B, Zhu K, et al.	2009–2025	3D Stacked IC Chip, 2.5D/3D Chiplet Heterogeneous System.	Thermal through-silicon via (TTSV) can reduce the interlayer thermal resistance of 3D IC by 40%; the co-design of microfluidic and TSV reduces the thermal resistance by 50% compared with the single scheme; the thermal simulation accuracy of the chiplet system is 90% higher than that of the traditional model.
Chip Thermal Characterization and Temperature Measurement Technology	Sarua A, PRAKASH S, et al.	2006–2024	GaN HEMT Power Transistor.	

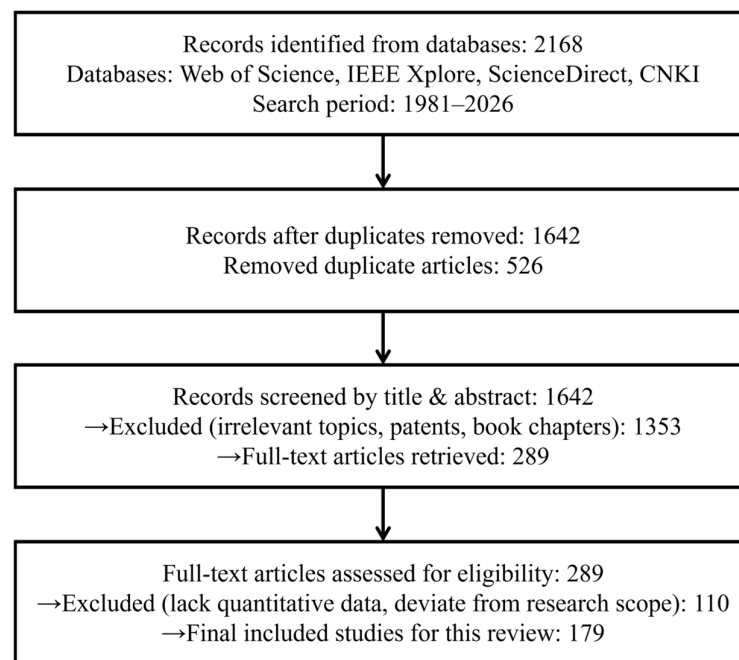


Figure 1. PRISMA flow diagram illustrating the literature retrieval and screening process of this review.

2. Heat Generation Characteristics and Thermal Challenges of High-Power Integrated Processors

2.1. Dual Rise in Power Consumption and Heat Flux Density

For a long time, the semiconductor industry has relied on the synergy of Moore’s Law and Dennard Scaling to maintain relatively stable power density while increasing transistor integration. However, as feature sizes approach physical limits, Dennard Scaling has effectively broken down since the 22 nm node, resulting in power density no longer decreasing with process scaling [10]. This fundamental shift has transformed the heat generation issue of high-performance computing chips from a “tolerable byproduct” into the primary bottleneck restricting performance scaling.

The exponential growth in computing demand in the era of artificial intelligence has further exacerbated this contradiction. The power consumption of mainstream AI accelerators has increased significantly, from approximately 400 W for the NVIDIA A100 to 700 W for the NVIDIA H100, while the AMD MI300X reaches a power envelope of about 750 W [11]. However, single-chip TDP is only one aspect of the problem. The real challenge lies in the fact that this high power is confined to an increasingly small chip area, leading to a sharp rise in heat flux density. For example, the local hotspot heat flux density of modern AI chips can now exceed 1 kW/cm², far exceeding the ultimate load-bearing capacity of traditional air cooling [2].

To complicate matters further, to break through the performance bottleneck of monolithic chips, the industry has shifted from monolithic system-on-chips (SoCs) to 2.5D/3D chiplet heterogeneous integration architectures. While this architecture offers significant advantages in bandwidth, yield, and intellectual property (IP) reuse, it also introduces unprecedented thermal coupling problems [12]. Feng et al. [12] pointed out through lumped-parameter models and finite element simulations that in 2.5D integration, significant thermal interaction paths are formed between adjacent chiplets through-silicon interposers, micro-bumps, and redistribution layers (RDLs). When an active chip operates, the heat it generates is laterally conducted to adjacent passive chips through the high-thermal-conductivity silicon interposer, causing the latter to experience a significant

temperature rise even without self-heating. They further found that as the chiplet pitch decreases from 0.5 mm to 0.1 mm, the maximum temperature rise of the passive chip can increase by nearly three times. This means that in the pursuit of high-density integration, thermal interaction problems will not be automatically resolved but will become increasingly severe.

Early research optimistically believed that using high-thermal-conductivity thermal interface materials or enhanced convective heat transfer could effectively isolate thermal effects between chips. However, scholars Virmani and Chatterjee noted [3] that this single-point optimization approach ignores the non-linear characteristics of multi-scale thermal coupling. Their analysis shows that in 3D stacked structures, multiple bonding interfaces, through-silicon vias (TSVs), and underfill layers form complex anisotropic heat conduction paths, making it often insufficient to suppress overall thermal interaction by simply reducing the thermal resistance of a single part. On the contrary, excessively increasing the thermal conductivity of the interposer may instead exacerbate thermal coupling between hotspots, leading to more uneven temperature distribution. This counterintuitive finding challenges the traditional design philosophy that “higher material thermal conductivity is always better”, revealing that thermal management in chiplet systems must be re-examined from the perspective of system-level, architecture-cooling co-design.

It can thus be seen that the heat generation problem of high-power AI chips has evolved from “single-point high temperature” to a systematic challenge of “multi-core interactive thermal fields”. The dual rise in power consumption and heat flux density, coupled with the increasingly tight thermal coupling between chiplets, has rendered traditional thermal management methods increasingly ineffective. Subsequent chapters will further explore systematic solutions such as advanced cooling technologies based on this foundation.

To place the above qualitative observations on a quantitative footing, we recall two fundamental heat transfer equations that govern thermal behavior at the chip level. For heat conduction within solid materials such as the silicon die and packaging layers, Fourier’s law states that

$$q = -k \frac{dT}{dx} \quad (1)$$

where q is the heat flux (W/m^2), k is the thermal conductivity ($\text{W}/(\text{m}\cdot\text{K})$), and dT/dx is the temperature gradient. Given that the thermal conductivity of silicon is only approximately $150 \text{ W}/(\text{m}\cdot\text{K})$, a local heat flux of $1 \text{ kW}/\text{cm}^2$ ($10^7 \text{ W}/\text{m}^2$) would require an extremely steep temperature gradient across the chip thickness, directly explaining the formation of severe hotspots. This is the physical reason why substrate conduction alone is inadequate and why high-thermal-conductivity materials or shortened conduction paths are mandatory. For heat removal from the chip surface to a coolant, Newton’s cooling law defines the convective heat flux as

$$q = h(T_s - T_f) \quad (2)$$

where h is the convective heat transfer coefficient ($\text{W}/(\text{m}^2\cdot\text{K})$), T_s is the chip surface temperature, and T_f is the fluid temperature. Under forced air cooling, h typically ranges from 50 to $250 \text{ W}/(\text{m}^2\cdot\text{K})$. To dissipate $300 \text{ W}/\text{cm}^2$ ($3 \times 10^6 \text{ W}/\text{m}^2$) with such a low h , the required temperature difference ($T_s - T_f$) would reach thousands of kelvin, making air cooling physically impractical. In contrast, single-phase liquid cooling can provide h values on the order of 10^3 – $10^4 \text{ W}/(\text{m}^2\cdot\text{K})$, reducing the required temperature difference by an order of magnitude. These two equations thus establish the quantitative baseline for understanding why traditional thermal management methods are no longer adequate for high-power AI chips, and they justify the advanced liquid cooling and packaging-level solutions to be discussed in the following sections.

2.2. Thermal Management Complexity Brought by Chiplet and 3D IC Architectures

Chiplet and 3D IC architectures break the performance and area bottlenecks of traditional monolithic chips through heterogeneous integration and vertical stacking, and have become the mainstream technical route for high-power AI chips. Nevertheless, such architectural innovation also brings unprecedented thermal management challenges, whose complexity is continuously exposed and aggravated with the improvement of integration density.

As early as 2014, Garrou et al. [13] systematically reviewed 3D integration technologies and pointed out that through-silicon vias, micro-bumps and multi-layer interfacial materials in vertically stacked structures form a high-resistance heat transfer network. This greatly increases the thermal resistance between layers, making it difficult for heat in the core layers to dissipate to external heat sinks, and, thus, exacerbating internal thermal gradients and hotspot issues within chips. This finding first revealed the inherent bottlenecks of 3D integration architectures in heat transfer paths, laying an important theoretical foundation for the subsequent thermal design of chiplets and high-power AI chips, and also alerting the industry to thermal risks at the architectural level.

With the large-scale deployment of 2.5D/3D chiplet architectures in AI processors, new thermal problems have gradually emerged. In 2024, Chen et al. [14] further discovered prominent interlayer thermal coupling effects in these architectures. As shown in Figure 2, dynamic loads intensify the transient thermal gradients inside chips. Heat from local hotspots spreads rapidly through vertically stacked layers, which not only raises the risk of thermal stress failure of interconnect structures, but also poses greater challenges to the design of dynamic thermal management strategies. Compared with the static thermal resistance issues proposed by Garrou, the research by Chen et al. extends the research scope to thermal behaviors under dynamic loads, and demonstrates the limitations of conventional static thermal design methods when applied to high-power AI chips.

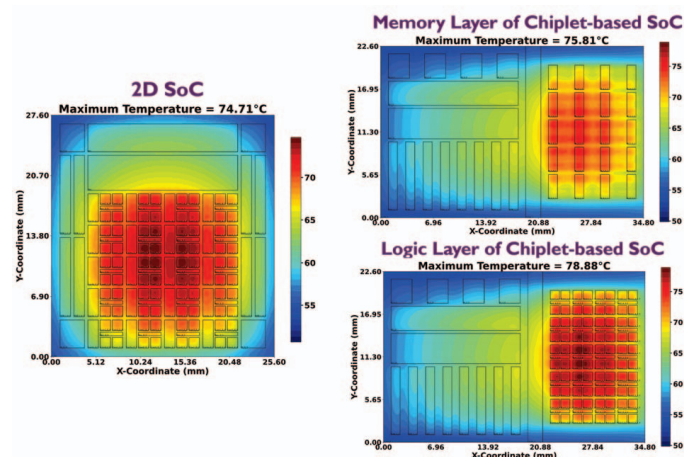


Figure 2. Steady-state thermal distribution of traditional 2D SoC, 2.5D and 3D chiplet packaging (reproduced with permission) [14].

In the same year, Yoon et al. [15] conducted a study on multi-chiplet packaging for large-scale AI neural processing units (NPUs) and further supplemented the thermal challenges induced by heterogeneous integration from the horizontal perspective. As illustrated in Figure 3 by Garrou et al., chiplets with different functions in heterogeneous integration systems feature distinct power consumption, which readily gives rise to local hotspots with extremely high heat flux density. Meanwhile, cross-scale packaging structures further increase the difficulty of thermal modeling. This finding improves the understanding of thermal issues in chiplet architectures. Apart from thermal resistance and thermal coupling

in the vertical direction, uneven power distribution and thermal crosstalk in the horizontal direction also stand as core challenges for thermal management.

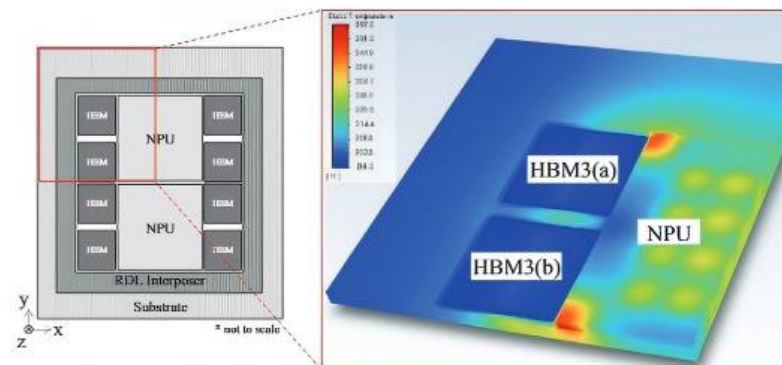


Figure 3. Temperature contour of NPU+HBM multi-chiplet packaging (obvious local hotspots formed in the NPU region) (reproduced with permission) [13].

In 2026, the research conducted by Zou Qifan's team further advanced the understanding of the complex thermal management issues in chiplet and 3D IC architectures. They pointed out [16] that after adopting chiplet and 3D integration solutions for high-power AI chips, horizontal thermal crosstalk becomes increasingly severe and further deteriorates the overall temperature uniformity. Such horizontal thermal coupling across chiplets causes localized hotspots to spread to adjacent chiplets, creating extensive thermal gradients and greatly increasing the difficulty of thermal management. From vertical thermal resistance and interlayer thermal coupling to horizontal thermal crosstalk, academic understanding of thermal issues in chiplet architectures has continued to deepen, which also demonstrates the inadequacy of conventional heat dissipation technologies for new-generation high-power AI chips.

Overall, research on thermal management of chiplet and 3D IC architectures has evolved progressively, from recognizing the bottleneck of static thermal resistance to identifying dynamic interlayer coupling, and further to exploring horizontal thermal crosstalk across chiplets. These studies reveal that the thermal challenges of high-power AI chips are not limited to one-dimensional heat dissipation constraints, but constitute systematic problems arising from vertical stacking, heterogeneous integration and dynamic workloads. Traditional thermal design methods for monolithic chips can no longer meet the requirements of emerging architectures. Accordingly, future research on heat dissipation technologies must break conventional thinking. From the perspective of cross-scale and multi-physics field coupling, integrated thermal management solutions tailored for chiplet and 3D IC architectures need to be developed.

Heterogeneous integration of multi-chiplet architectures induces multi-dimensional thermal coupling effects across vertical interlayers and lateral planes. Conventional steady-state thermal simulations tend to underestimate device aging risks originating from dynamic thermal stresses. Several researchers have investigated active thermal control strategies for SiC power devices, wherein transient chip temperature fluctuations are suppressed via dynamic gate modulation. This technique can reduce the amplitude of thermal cycles by over 30%, substantially mitigating delamination and bonding failures induced by thermal mismatch. Such design philosophy is equally transferable to the dynamic thermal management of multi-chiplet AI processors [17].

2.3. Limitations of Air Cooling and the Necessity of Liquid Cooling for High-Power Integrated Processors

Air cooling uses air as the heat exchange medium. Thanks to its simple structure, low deployment cost and convenient operation and maintenance, it has long been the dominant heat dissipation solution for conventional chips. Nevertheless, with the continuous growth of power consumption of high-power AI chips and rack power density, the performance bottlenecks of this traditional approach have become increasingly prominent. Accordingly, liquid cooling has become an inevitable choice to address heat dissipation challenges under high heat flux density.

2.3.1. Drawbacks of Air Cooling for High-Power Integrated Processors

Air cooling is first constrained by its physical limit in heat dissipation capacity. As reported by Crosley et al. [18], limited by the intrinsic thermal conductivity of air, the maximum heat dissipation capacity of a single rack with conventional air cooling is merely 15 to 20 kW. Currently, the power consumption of a single high-power AI chip ranges from 700 W to 1500 W, and the power density of a single rack exceeds 50 kW. Such values far surpass the bearing capacity of air cooling, and obvious chip frequency throttling will occur once the load exceeds the limit. These figures clearly illustrate the huge gap between air cooling performance and the computing demands of AI systems.

Apart from insufficient heat dissipation capacity, the drawbacks of air cooling in terms of energy consumption and operational stability have also become evident. According to the research conducted by Zou Qifan et al. [16], in data centers with high power density, the power consumption of air cooling fans increases with the cube of wind speed, which substantially raises the overall energy consumption of the facility. Figure 4 compares the energy consumption composition of data centers adopting traditional air cooling and liquid cooling. For air-cooled data centers, cooling accounts for 43% of the total server power consumption, while the proportion drops to only 9% in liquid-cooled ones, which fully reflects the extra energy burden brought by air cooling [16]. Meanwhile, the study also demonstrates [16] that chips undergo large temperature fluctuations under air cooling, making long-term stable operation difficult to guarantee. Figure 5 presents the temperature comparison of GPU chips under different cooling solutions. With the same inlet temperature, the GPU temperature under air cooling is considerably higher than that under liquid cooling. As the inlet air temperature rises, the chip temperature increases sharply. In contrast, liquid cooling maintains the chip temperature at a lower and more stable level. Suffering from both excessive energy consumption and poor stability, air cooling shows increasingly prominent deficiencies in high-power AI application scenarios.

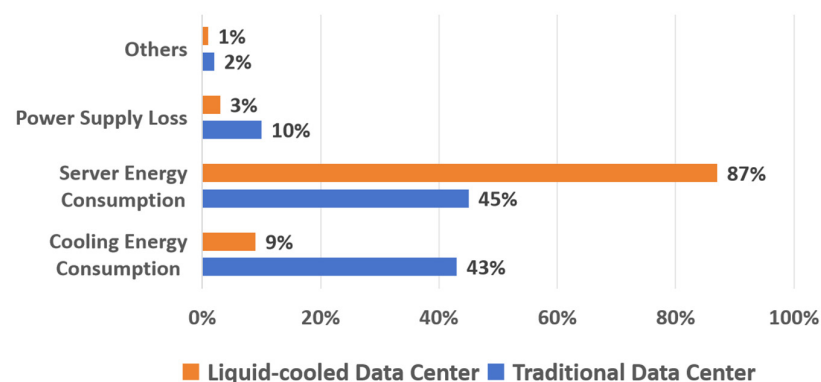


Figure 4. Energy consumption comparison between traditional air cooling and liquid cooling.

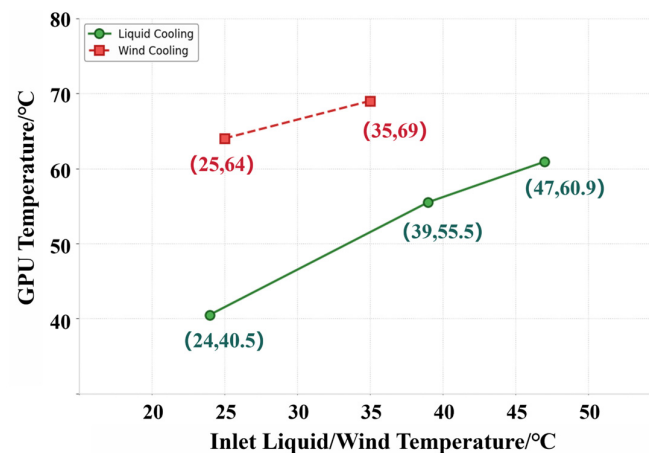


Figure 5. Temperature comparison of GPU chips under air cooling and liquid cooling schemes.

2.3.2. Liquid Cooling as a Critical Path to Breaking the Heat Dissipation Bottleneck

Faced with the overall limitations of air cooling, liquid cooling has emerged as the core solution for heat dissipation of high-power AI chips thanks to its superior heat transfer performance. In 2024, Azarifar M et al. [19] quantitatively compared the two technologies based on the fundamental principle of heat transfer coefficient. When the heat flux density of AI chips exceeds approximately 100 W/cm^2 , conventional air cooling gradually approaches its practical limit. Forced-air cooling typically provides heat transfer coefficients of about $50\text{--}250 \text{ W/(m}^2\cdot\text{K)}$, whereas single-phase liquid cooling can reach $10^3\text{--}10^4 \text{ W/(m}^2\cdot\text{K)}$, and two-phase cooling may further increase this value to $10^4\text{--}10^5 \text{ W/(m}^2\cdot\text{K)}$. Therefore, liquid cooling offers substantially higher heat transfer capability than air cooling and is a promising solution for high-heat-flux AI chip thermal management.

Liquid cooling boasts advantages not only in heat transfer efficiency, but also in chip thermal uniformity and long-term reliability. Patel A et al. [20] conducted research on 2.5D/3D stacked AI chips. The results show that the temperature difference at chip hotspots exceeds 25 K under air cooling, which is highly likely to cause local thermal failure. By contrast, liquid cooling limits the temperature difference within 5 K, greatly reducing the risk of interconnect failure induced by excessive thermal gradients and ensuring the long-term stable operation of chips.

In addition, liquid cooling delivers remarkable performance in cutting the overall energy consumption of data centers. According to the study by Jalili M et al. [21], the power usage effectiveness (PUE) of high-power AI chip clusters adopting air cooling is generally higher than 1.5, while immersion liquid cooling can reduce the PUE to below 1.1, effectively lowering the energy consumption and operational costs of data centers. From heat transfer performance and energy efficiency optimization to reliability assurance, liquid cooling fully compensates for the defects of air cooling and has become an inevitable choice for heat dissipation of high-power AI chips.

The comparative analysis in this section demonstrates that the strengths of air cooling in the era of low-power conventional chips no longer apply to high-power AI chips. Its heat dissipation capacity, energy efficiency and temperature stability fail to meet the demands of new-generation computing power. With a hundredfold improvement in heat transfer performance, excellent thermal uniformity and prominent energy-saving effects, liquid cooling presents irreplaceable advantages in both fundamental theory and engineering practice. It indicates that liquid cooling is not merely a supplementary solution to air cooling, but an important route to address thermal challenges of high-power AI chips. It also points out the direction for subsequent research on advanced liquid cooling technologies tailored for chiplet and 3D IC architectures.

The inferiority of air cooling relative to liquid cooling can be formally quantified through the series thermal resistance network that governs heat flow from the chip junction to the ambient environment:

$$R_{\text{total}} = R_{\text{jc}} + R_{\text{TIM}} + R_{\text{ca}} \quad (3)$$

where R_{jc} is the junction-to-case (or junction-to-substrate) conduction resistance, R_{TIM} is the thermal interface material resistance, and R_{ca} is the case-to-ambient (or case-to-coolant) convective resistance. In conventional air-cooled systems, R_{ca} dominates due to the low convective heat transfer coefficient of air. In liquid-cooled systems, however, the bottleneck shifts to R_{jc} and R_{TIM} , as the high h of liquid convection drastically reduces R_{ca} . This additive expression explains why near-junction cooling (targeting R_{jc}), advanced TIMs (targeting R_{TIM}), and high-performance cold plates (targeting R_{ca}) are not mutually exclusive alternatives but, rather, complementary components that must be co-optimized. The following section systematically reviews how each technology addresses a specific term in this resistance chain, ultimately achieving the substantial reduction in R_{total} required for high-power AI processors.

3. High-Efficiency Thermal Management Technologies

3.1. Near-Junction Liquid Cooling Technology

Near-junction liquid cooling technology is a high-efficiency thermal dissipation solution that introduces cooling media directly into the vicinity of the active junction surface of chips. Its core principle lies in fabricating microchannels within silicon substrates or packaging substrates, allowing coolant to flow directly through the near-source region, thereby rapidly removing heat via convective heat transfer, significantly shortening heat transfer paths and reducing interfacial thermal resistance. The prominent advantage of this technology is its proximity of heat exchange to the heat source, enabling it to efficiently dissipate ultra-high heat flux densities exceeding 300 W/cm^2 and mitigate local hotspot issues. Furthermore, it demonstrates strong process compatibility and represents one of the critical technologies that enable high-power AI processors to break through the bottlenecks of traditional thermal management schemes.

Various heat dissipation approaches have been proposed to suppress junction temperature rise of high-power chips, which can be grouped into passive and active cooling modes. A mature universal classification standard for power semiconductor cooling schemes was summarized by Górecki and Posobkiewicz in their comprehensive review, where all existing heat dissipation systems were systematically sorted according to driving energy consumption, heat transfer medium and packaging integration form [22]. The framework divides cooling technologies into passive schemes (heat sinks, heat pipes, phase-change materials, high-thermal-conductivity substrates) without auxiliary power supply, and active schemes (forced air cooling, single/two-phase liquid cooling, jet impingement cooling) dependent on pumps or fans.

This classification rule is adopted as the basic analytical framework of this chapter. On this basis, we further adjust the discussion focus, shifting from medium-low power discrete power devices in the original literature to ultra-high heat flux AI integrated chips, and analyze the performance bottlenecks of each cooling method under chip-level packaging constraints.

3.1.1. Microchannel Liquid Cooling

Microchannel cooling technology is a chip-level liquid cooling solution that involves fabricating micron-scale flow channels directly within or adjacent to the chip surface and

introducing coolant into these channels. By significantly shortening the heat transfer path and eliminating the thermal resistance bottlenecks inherent in traditional thermal dissipation systems—specifically those associated with silicon substrates and TIMs—it enables the coolant to directly remove heat via convective or phase-change heat transfer in extremely close proximity to the chip heat source, thereby achieving efficient thermal dissipation for high heat flux density chips.

Figure 6 illustrates the structure of embedded straight single-phase microchannels [23]. Straight through-type single-phase microchannels feature simple structures and low manufacturing costs. However, smooth channel walls tend to develop fully developed fluid boundary layers, which limits the heat transfer capacity.

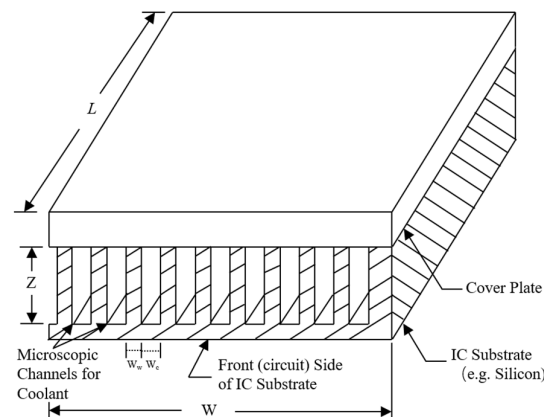


Figure 6. Schematic diagram of embedded straight microchannel (reproduced with permission) [23].

To quantitatively assess the convective performance of these straight single-phase microchannels, the Nusselt number is commonly estimated by the Dittus–Boelter correlation for turbulent internal flow:

$$Nu = 0.023Re^{0.8}Pr^{0.4} \quad (4)$$

where $Nu = hD_h/k_f$, $Re = \rho u D_h/\mu$ is the Reynolds number, and $Pr = c_p\mu/k_f$ is the Prandtl number of the coolant. Although originally developed for macro-scale tubes, this correlation provides a first-order design guideline for straight microchannels. It clearly demonstrates that reducing the hydraulic diameter D_h enhances h , but at the cost of a sharply increased pressure drop—scaling as approximately $\Delta P \propto 1/D_h^2$ in laminar flow. This intrinsic trade-off is precisely why the following ribbed and manifold structures are proposed: they aim to enhance heat transfer (increase Nu) without incurring an excessive pumping penalty, thereby improving the overall thermal–hydraulic performance beyond the limitation of plain straight channels.

In addition, long individual channels lead to a large temperature rise gradient of coolant along the flow direction, resulting in poor global temperature uniformity across multi-die chips.

To intensify fluid turbulence and expand the effective heat transfer area, microchannels with periodic ribs have been widely adopted (Figure 7) [24]. Compared with straight microchannels, rib structures can significantly enhance the heat transfer coefficient. However, ribs drastically increase fluid flow resistance, which substantially raises the pumping power consumption of the cooling system. In addition, such structures cannot address the axial temperature difference induced by long flow channels.

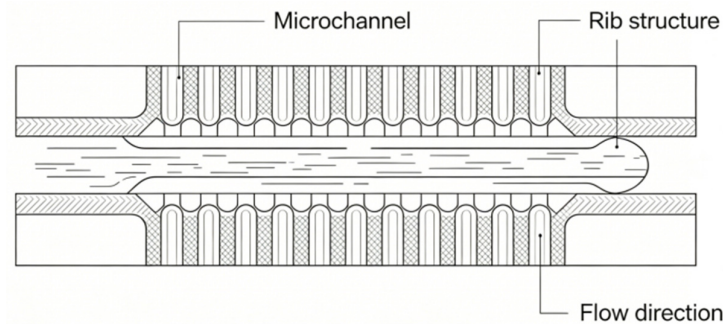


Figure 7. Schematic diagram of periodically ribbed microchannel structure.

Tang et al. [25] proposed a manifold microchannel structure with multiple inlets and outlets. This structure shortens the flow length of each single channel via multi-path flow splitting, balancing flow resistance and heat transfer performance while optimizing temperature uniformity. Figure 8 presents the three-dimensional schematic of an additively manufactured manifold microchannel heat sink (MMCHS). The left side shows the overall heat sink structure with fluid inlets and outlets, while the enlarged cross-section on the right labels the upper 5 mm manifold and lower 3 mm microchannels. Colored streamlines intuitively illustrate the internal flow path: the coolant sprays vertically into the microchannels for heat exchange before converging upward. Nevertheless, limited by the sensible heat transfer mechanism of single-phase working fluids, this structure still fails to satisfy the heat dissipation demands of computing chips with ultra-high heat flux.

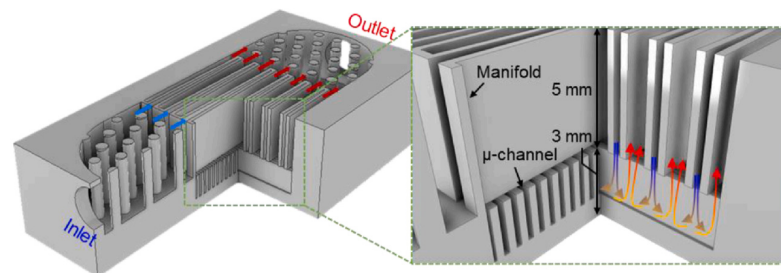


Figure 8. Three-dimensional structural schematic of additively manufactured MMCHS. The blue arrows indicate the incoming low-temperature coolant, while the red arrows represent the heated coolant after heat absorption. (reproduced with permission) [26].

Embedded microchannels arrange flow passages adjacent to the chip active region [27], minimizing the solid heat conduction path and reducing substrate thermal resistance to the greatest extent (Figure 9). Compared with the aforementioned external microchannels, they deliver prominent thermal resistance advantages. Yet, this structure faces high difficulties in wafer sealing and poor compatibility with conventional complementary metal–oxide–semiconductor (CMOS) manufacturing processes. It is only applicable to customized special chips, restricting its large-scale commercial deployment.

Single-phase microchannels have an inherent ceiling on heat transfer capacity. In contrast, two-phase microchannels can withstand higher thermal loads by leveraging the latent heat of working fluid phase change. Peterson's research group [28] conducted studies on silicon-based microgroove heat pipes, pioneering the research field of microscale phase-change heat transfer. Krishnamurthy and Peles [29] adopted a micro-pin fin array structure to increase the density of boiling nucleation sites and elevate the critical heat flux (CHF). Later, Palko's team [30] proposed a diamond composite porous two-phase microgroove structure, as illustrated in Figure 10. Relying on the high lateral thermal conductivity of

diamond and boiling nucleation enhancement on porous surfaces, this structure can handle kilowatt-level local hotspots.

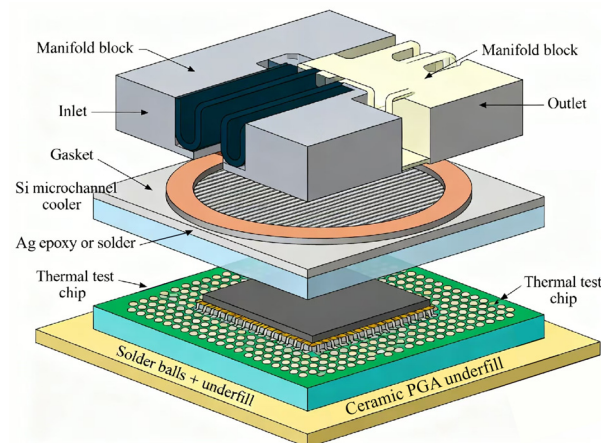


Figure 9. Schematic cross-section of microchannel cooler integrated in a single chip module. Different colors represent ceramic substrate, silicon chip, and microchannel cooling layer, respectively.

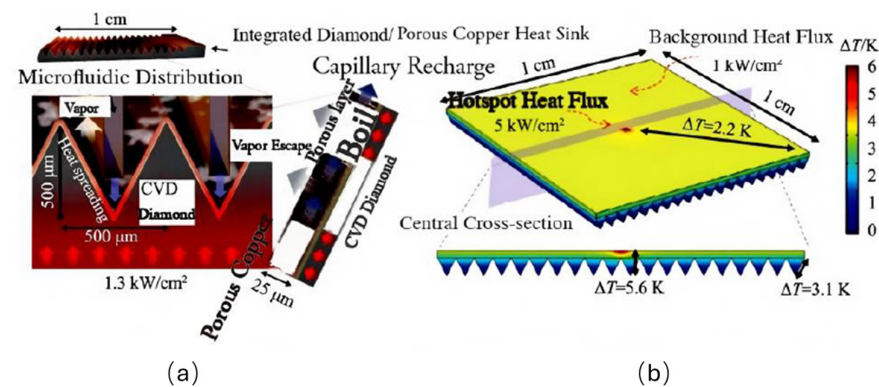


Figure 10. Composite diamond-porous copper microgroove two-phase cooling structure for hotspot thermal management (reproduced with permission) [30]. (a) Schematic diagram of microfluidic liquid supply, capillary replenishment and vapor escape inside the microgroove unit with a heat flux of 1.3 kW/cm². (b) Temperature rise ΔT distribution nephogram of the 1 cm × 1 cm chip under combined heat loads (5 kW/cm² central hotspot and 1 kW/cm² background heat flux); the color bar on the right corresponds to the magnitude of temperature rise ΔT in Kelvin (K), and the cross-section displays the temperature distribution of the central plane.

The heat dissipation limit of such two-phase microchannel structures is ultimately constrained by the critical heat flux—the maximum heat flux beyond which the vapor film insulates the heated surface, causing a sudden and drastic temperature rise. Among various predictive models, the Kandlikar correlation, which explicitly accounts for channel geometry, fluid properties, and flow quality, is widely adopted for microchannel boiling analysis:

$$q_{CHF} = h_{fg} G \left[\frac{We}{We + C} \right]^{0.5} (1 - x_{in}) \quad (5)$$

where h_{fg} is the latent heat of vaporization, G is the mass flux (kg/(m²·s)), $We = G^2 D_h / (\rho_l \sigma)$ is the Weber number characterizing the inertial-to-surface-tension force ratio, x_{in} is the inlet vapor quality, and C is an empirical constant depending on channel geometry. This correlation reveals that CHF is strongly dependent on mass velocity and vapor quality. In practical terms, it explains why the diamond composite porous structure was designed not only to enhance nucleation site density but also to delay CHF onset through improved

liquid supply to the heated surface. Similarly, the membrane-based micro-evaporator discussed below addresses CHF by enabling efficient vapor removal, thereby postponing the vapor blanketing effect that triggers CHF. Collectively, these two-phase enhancements push the practical heat dissipation limit beyond 1 kW/cm^2 —a requirement that single-phase microchannels alone cannot meet.

Unfortunately, diamond substrates suffer from high material costs and high processing barriers, making mass commercialization impractical. Membrane-based micro-evaporators (Figure 11) realize automatic gas–liquid separation via hydrophobic membranes [14] and exhibit outstanding heat dissipation performance under low-pressure operating conditions, with cost advantages over diamond composite structures. However, the membranes tend to degrade and fail after long-term operation at high temperatures, leading to insufficient engineering reliability.

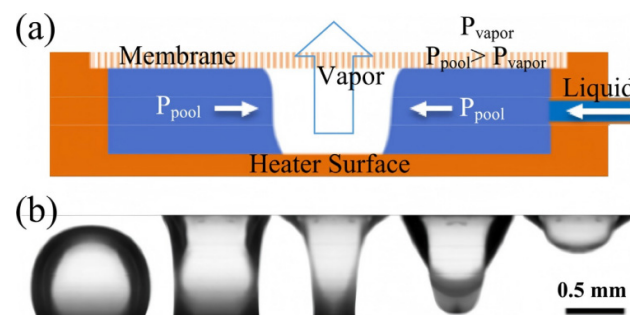


Figure 11. (a) Schematic cross-section of the membrane-based heat sink (not to scale) and (b) visualization of an air bubble extraction process from a pool of water constrained by a hydrophobic membrane in an adiabatic test. Reprinted/adapted from Fazeli and Moghaddam under CC BY 4.0. [31].

Both straight single-phase and two-phase microchannels suffer from excessively long single-channel flow paths and uneven flow distribution, making them incompatible with the non-uniform heat generation characteristics of chiplet and 3D-stacked chips. Manifold microchannels can effectively resolve this bottleneck [32]. Basic double-layer manifold microchannels improve flow distribution through two-stage liquid splitting on the upper and lower layers (Figure 12). Hierarchical array two-phase manifolds add multi-stage liquid distribution structures [33] to shorten the heat exchange travel of coolant and drastically reduce overall flow pressure drop (Figure 13). Three-dimensional embedded manifolds realize multi-path liquid distribution within three-dimensional space [34]. Compared with planar manifolds and traditional straight cold plates, they can achieve higher heat dissipation power under low pumping power consumption (Figure 14). Chip-flow channel integrated manifold structures eliminate additional thermal resistance at bonding interfaces, delivering superior heat dissipation efficiency to separately fabricated manifolds. However, the integrated forming process is complex with low production yield.

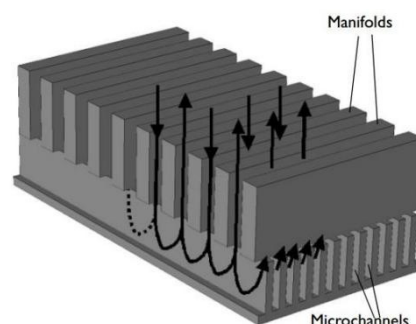


Figure 12. Schematic diagram of manifold microchannel heat sink structure.

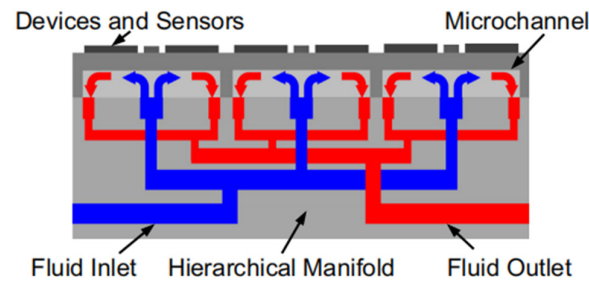


Figure 13. Cross-sectional schematic diagram of the hierarchical manifold microchannel heat sink array design concept (reproduced with permission) [33].

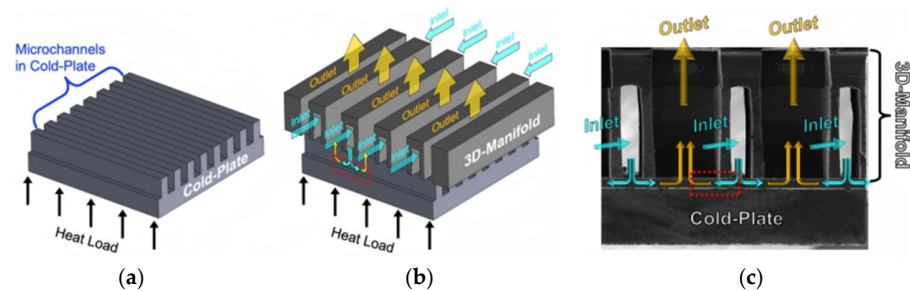


Figure 14. Evolution schematic of microchannel cooling architectures. (a) Conventional parallel microchannel cold plate under bottom heat load; (b) 3D manifold integrated microchannel cooling configuration with distributed inlet/outlet manifolds; (c) Cross-sectional micrograph of the manifold-microchannel assembly, annotated with fluid inlet and outlet flow directions (reproduced with permission) [34].

Recent research has further enriched manifold microchannel designs. Wu et al. [35] proposed a leaf-vein-inspired fractal manifold structure combined with intelligent thermal optimization, achieving synergistic improvements in temperature uniformity and pressure drop for chiplet-based AI processors. An embedded manifold microchannel solution that maintains high heat transfer coefficients while simplifying packaging integration has also been demonstrated [36]. Concurrently, a unified parametric optimization framework for microchannel fin geometries was developed, enabling rapid selection of optimal fin shapes, spacings, and heights—a valuable complement to empirical design approaches [37].

Considering the performance, cost and process constraints of all the above microchannel solutions, differentiated selections can be made for various chip application scenarios: (a) Ribbed and split-flow single-phase microchannels feature mature manufacturing processes and high cost performance, suitable for general integrated chips with low-to-medium power consumption; (b) Diamond composite two-phase microgrooves deliver the maximum heat transfer limit, ideal for high-end computing chips with prominent local hotspots, yet their mass production costs remain prohibitively high; (c) Embedded microchannels offer the lowest thermal resistance, and are only available for customized special military chips; (d) Three-dimensional hierarchical manifold microchannels provide uniform flow distribution and low flow resistance, currently serving as a great microchannel heat dissipation solution for chiplet and three-dimensional stacked AI chips.

3.1.2. Direct Jet Impingement Cooling

Direct jet impingement cooling is an active heat transfer technology with prominent engineering merits for high heat flux cooling of high-power AI processors. This technology adopts high-precision nozzle arrays to jet low-temperature coolant toward the chip heat source surface in a vertical impingement manner, rapidly stripping and breaking the laminar thermal boundary layer adjacent to the device wall. Unlike passive cooling schemes

such as air cooling and conventional microchannel heat transfer, jet impingement cooling is free from the constraint of solid-wall conduction thermal resistance. It delivers an extremely high local convective heat transfer coefficient and outstanding heat flux bearing capacity, enabling targeted suppression of localized chip hotspots and rapid heat evacuation. Accordingly, it has become a mainstream enhanced heat transfer solution to address the cooling bottleneck of concentrated, non-uniform ultra-high heat flux in AI chips.

Reviewing the structural evolution of jet impingement cooling, the overall technical iterations have been continuously optimized around four core engineering challenges: suppressing crossflow interference, homogenizing full-domain flow distribution, mitigating cross-regional thermal crosstalk, and balancing cooling capacity with pumping power consumption. Each generation of structural upgrades systematically remedies the drawbacks of prior configurations, including uneven fluid distribution, non-uniform heat transfer mechanisms and excessive flow resistance loss, gradually adapting to the cooling demands of large-size, high-power advanced AI computing chips with non-uniform heat generation.

Single discrete nozzles represent the fundamental original configuration of jet cooling, featuring separate scattered nozzle layouts that generate high-intensity localized impingement heat transfer via single-hole directional jets. Its core technical strengths lie in concentrated jet energy, intense targeted heat transfer and simple, integrable structures. It can rapidly suppress extreme hotspots over tiny chip areas, making it suitable for emergency cooling scenarios with transient localized ultra-high heat flux. Nevertheless, discrete nozzles carry inherent structural flaws from the perspective of full-domain heat transfer mechanisms. Their discrete arrangement fails to form continuous, uniform coverage of impingement heat transfer, which cannot meet the full-domain temperature uniformity control requirements of large-size AI chips. More importantly, multiple independent jets generate radially diverging fluid upon striking the wall. Divergent flows from different regions converge and overlap to form large-scale secondary crossflow. Such crossflow sweeps subsequent incoming primary jets horizontally along the chip surface, distorting jet impingement profiles and weakening stagnation-point impact intensity. This drastically reduces heat transfer efficiency at chip edge zones, resulting in an enlarged temperature gap between the chip center and edges and degraded full-domain temperature uniformity. Consequently, this configuration fails to satisfy the precise thermal management requirements of highly integrated AI processors.

To address the poor full-surface coverage and insufficient temperature uniformity of discrete nozzles, the silicon-based integrated micro-nozzle array scheme was first proposed and gradually applied in engineering practice [38]. As illustrated in Figure 15, this integrated assembly mainly consists of an upper liquid-collecting pressure stabilization cavity, a middle uniformly distributed micro-nozzle array, and a lower heated chip substrate coupled together. It realizes synchronous multi-point vertical impingement via micron-scale nozzle arrays, fundamentally overcoming the limitation of narrow coverage for single nozzles. This configuration achieves full-coverage impingement heat transfer over the chip surface through standardized array layout, significantly improving the continuity and stability of cooling zones. Meanwhile, silicon micromachining technology enables precise control of critical geometric parameters including nozzle diameter, nozzle pitch and impingement height, which effectively enhances the overall heat transfer uniformity and structural integration level of the chip. Relevant parametric studies [39] systematically investigated the coupled effects of core structural parameters (nozzle diameter, array pitch, jet impingement distance, etc.) on convective heat transfer coefficient, wall thermal resistance and pumping pressure drop via a multi-objective numerical optimization framework. Standardized design criteria for regular array jets were established to reasonably balance basic cooling performance and fluid pumping loss. Although regular array structures

greatly improve full-domain heat transfer capacity, they still adopt a flow layout with single-cavity unified liquid inlet and unified liquid collection [39], where waste fluid from all impingement nozzles shares one common collecting channel. Such centralized liquid collection causes massive high-temperature waste fluid to continuously accumulate and spread horizontally over the chip surface, forming a persistent intense crossflow field. This crossflow continuously entrains and disturbs the subsequent incoming cold fluid, degrading jet impingement performance in remote regions. As a result, crossflow interference cannot be fundamentally eliminated from the mechanism perspective, and obvious local heat transfer attenuation and temperature maldistribution still occur under operating conditions of large-size, high-power AI chips.

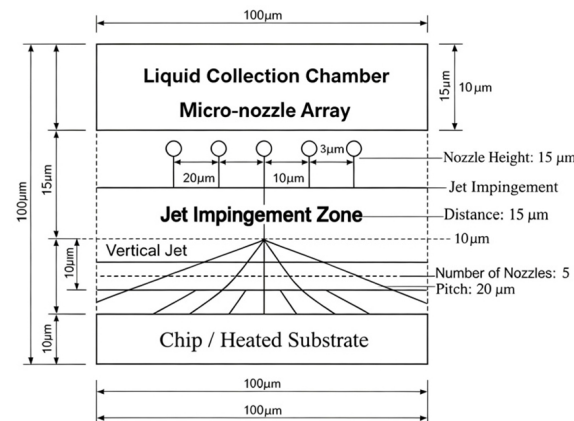


Figure 15. Schematic cross-section of the microjet impingement cooling structure, with marked structural dimensions including liquid collection chamber, micro-nozzle array, jet impingement zone and heated chip substrate.

On the basis of regular array structures, iterative advances in integrated molding technology further improve the engineering practicability of jet cooling modules. The monolithic multi-nozzle jet cooling module fabricated via 3D printing integration [40] abandons the split-chip manufacturing mode of traditional silicon etching. It enables integrated molding of the nozzle array, manifold chambers and sealing structures, as shown in Figure 16, which greatly simplifies the system assembly process and eliminates sealing leakage risks and assembly errors inherent to conventional split structures. Under operating conditions of a typical 64 mm² computing chip with a rated power consumption of 50 W, this integrated jet system delivers ultra-high heat flux cooling capacity of 78 W/cm². It significantly reduces interfacial thermal resistance while effectively restraining fluid pumping power loss, outperforming traditional silicon-based micro-nozzle arrays in both comprehensive cooling performance and engineering adaptability. Nevertheless, from the perspective of flow field mechanism, this optimization scheme only upgrades the manufacturing process without modifying the core flow path structure with centralized liquid collection. The inherent flow field defects including lateral accumulation of waste fluid on the chip surface and jet performance attenuation at remote zones remain unresolved. Therefore, this structure fails to match the complex heat generation characteristics of large-size AI chips featuring non-uniform and gradient heat distribution.

To address the core mechanical drawbacks of traditional single-cavity liquid collection structures, including severe crossflow accumulation, untimely drainage of waste fluid and deteriorated heat transfer at remote zones, the distributed jet configuration with zoned alternating inlet and outlet channels has emerged as a new-generation advanced jet cooling solution suitable for high-power AI processors [41]. This configuration thoroughly reconstructs the flow channel topology of conventional arrays by replacing the full-domain

unified liquid collection mode with an interleaved flow layout featuring independent liquid supply and drainage for separate zones. As illustrated in Figure 17, compared with the traditional Type-A unified liquid collection structure, the innovative Type-B zoned drainage structure guides and discharges impingement waste fluid from each sub-region independently and rapidly. It physically blocks the lateral spread and cross-zone sweeping of high-temperature waste fluid, drastically mitigating the distortion and suppression of primary jet profiles induced by crossflow fields, which effectively resolves the problems of weakened heat transfer at chip edges and excessive overall temperature discrepancy in large-scale chips. This structural optimization not only greatly improves flow field uniformity and heat transfer stability, but also establishes correlation models for Nusselt number and flow pressure drop applicable to distributed jet systems through extensive experimental tests under various operating conditions. The models accurately reveal the evolution laws of convective heat transfer and fluid resistance characteristics inside complex zoned flow channels, providing comprehensive theoretical support for parametric design, performance prediction and structural optimization of new-generation jet arrays adapted to large-size chips with non-uniform heat sources.

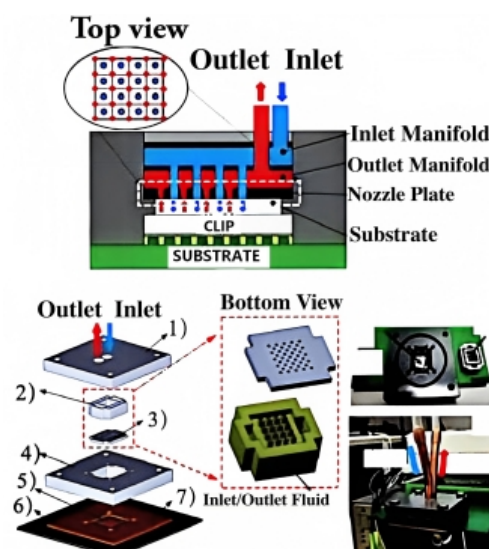


Figure 16. The array jet impingement cooling structure (under the Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License) [40].

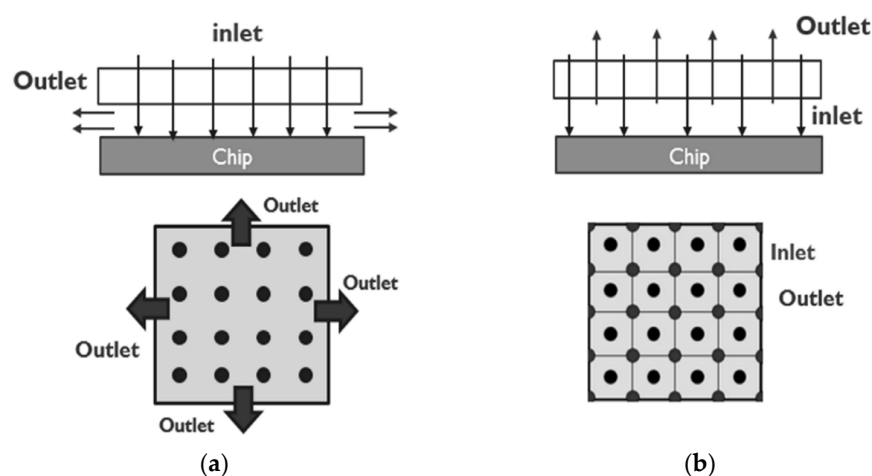


Figure 17. Impingement jet cooling configurations. (a) Configuration A: common outlets; (b) configuration B: distributed outlets (reproduced with permission) [41].

A horizontal comparison of the technical characteristics of the three mainstream jet configurations from the perspectives of technical mechanism and engineering adaptability leads to clear conclusions. Single discrete jet structures deliver concentrated heat transfer and feature simple architectures, yet they suffer from extremely poor full-chip adaptability and are only applicable to emergency targeted cooling for tiny localized hotspots on chips. Regular, equally-spaced micro-nozzle arrays exhibit favorable machinability and uniform full-surface coverage, capable of meeting the moderate heat flux cooling demands of general computing chips with even heat generation for standardized, universal board cooling scenarios. Nevertheless, their centralized liquid collection mechanism fundamentally prevents complete elimination of crossflow interference. By contrast, distributed jets with zoned alternating inlet and outlet channels rely on core mechanisms of zoned fluid guidance, waste fluid isolation and crossflow suppression to thoroughly resolve coupled flow field interference in conventional arrays. This configuration is well suited for complex operating conditions of high-power AI processors characterized by large chip sizes, ultra-high heat flux and non-uniform gradient heat generation, rendering it a great jet thermal management solution for high-end AI chips at present. Overall, the technical iteration of jet impingement cooling essentially centers on suppressing flow field interference, improving adaptability to diverse heat sources, and optimizing the trade-off between pumping power consumption and cooling performance. Follow-up technical innovations can be further advanced in fields including composite enhanced heat transfer design, intelligent adaptive regulation of zoned flow rates, and multi-scale coordinated flow field optimization, so as to provide technical support for extreme thermal management of ultra-high computing power AI chips.

3.1.3. Cold Plate Liquid Cooling

Cold plate liquid cooling attaches a metal cold plate with embedded sealed flow passages to the outer surface of chip packages. Thermal interface materials fill the micron-scale assembly gaps between the cold plate and packaging. Heat generated by the chip first transfers to the cold plate substrate via solid conduction, then waste heat is removed through convective heat exchange with internal circulating coolant. This cooling solution requires no modifications to the original chip manufacturing process and is compatible with all types of server hardware architectures. It features easy assembly and post-operation maintenance with controllable overall deployment costs, making it the most widely adopted mature liquid cooling technology for mass deployment in current data centers. Nevertheless, stacked multi-layer interfaces create series contact thermal resistance. Restricted by the inherent bottleneck of interfacial thermal resistance, the ultimate heat dissipation capacity of cold plates is inferior to near-junction microchannels and jet impingement cooling. Cold plates can only meet conventional steady-state cooling demands of mainstream high-power AI chips and fail to handle extreme hotspot conditions with local heat flux exceeding kilowatts per square centimeter. Four typical configurations have been developed for existing cold plate flow channel systems. Each configuration is optimized differently around three core trade-offs: flow pressure drop, full-chip temperature uniformity, and manufacturing difficulty, to match the cooling requirements of various computing scenarios.

Continuous serpentine channel cold plates represent the basic general-purpose configuration [42]. A complete S-shaped bent flow channel runs through the entire heat exchange area of the cold plate. Multiple bends alter the fluid flow direction and continuously disturb the near-wall fluid boundary layer, mitigating severe temperature rise of coolant along single straight channels and excessive temperature difference across the chip front and rear. The structure boasts simple overall manufacturing processes and low mass production costs, suitable for cooling small-to-medium power general servers. Figure 18 presents

the simulation schematic of a serpentine cold plate, which clearly marks the inlet and outlet of hot and cold fluids as well as the region where chip thermal load is applied, intuitively demonstrating the layout of continuous bent channels. However, this configuration has notable hydrodynamic drawbacks: numerous successive bends generate persistent vortex losses, drastically increasing fluid pressure loss along the flow path and raising power consumption of the supporting circulation pump. It cannot simultaneously achieve low pumping power and uniform temperature control across the whole plate, resulting in low comprehensive energy efficiency under long-duration full-load operation of high-density cabinets.

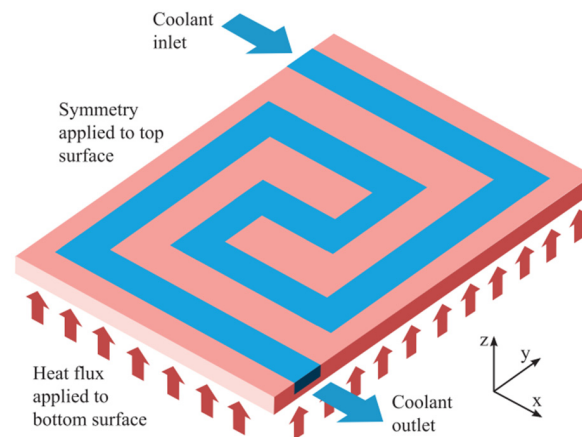


Figure 18. Schematic diagram of basic structure of serpentine flow channel cold plate (reproduced with permission) [42].

Internally inclined rib-reinforced cold plates are structurally optimized to address the high vortex loss and insufficient heat transfer coefficient of plain smooth serpentine channels. Arrays of inclined heat transfer ribs are arranged inside the cold plate. These ribs physically break the continuous fluid boundary layer and greatly expand the effective convective heat transfer area, improving heat transfer performance even at low coolant flow rates to compensate for the weak cooling capacity of smooth straight channels. This design fits standard medium-power AI accelerator cards [43]. Still, inclined rib arrays with fixed angles and layouts lack adaptive flow regulation capabilities. When matched with large-size integrated chips, fluid continuously dissipates energy along the main flow direction, leading to insufficient coolant supply in remote chip zones and persistent local hotspots at corners. Hence, this structure cannot resolve maldistributed flow across large-area chips.

To accommodate the non-uniform heat generation characteristics of high-power integrated chips, Patil et al. [44] quantified the influence of operating parameters on cold plate temperature control performance through variable flow rate and variable thermal load experiments, providing experimental references for parameter selection of cold plate operation. However, their channel layout still adheres to the conventional single-inlet single-outlet design. On this basis, grid split-flow cold plates adopt a grid liquid distribution layout with one inlet and multiple outlets [45]. The single main channel is split into numerous parallel branches that uniformly cover the entire heat exchange plane. The multi-path split-flow structure evenly distributes coolant flow to alleviate the inherent issues of insufficient fluid supply in remote areas and local overheating. It is inherently compatible with the fragmented non-uniform heat generation of chiplet multi-die architectures, greatly expanding the application scope of cold plate liquid cooling for advanced packaged computing chips. Figure 19 illustrates the schematic diagram of the grid-channel cold plate system. Compared with serpentine and inclined rib cold plates, the grid configuration

delivers prominent advantages in temperature uniformity. Yet, traditional mechanical milling and drilling processes struggle to fabricate intricate fine grid channels, leading to longer manufacturing cycles, reduced finished product yield, and noticeably higher costs for mass production.

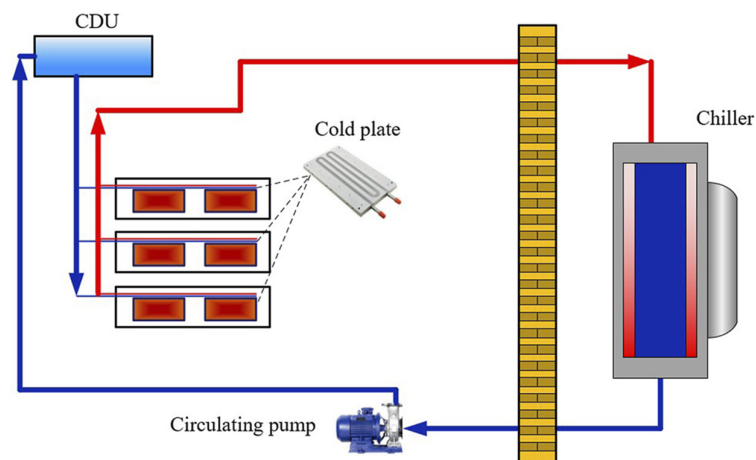


Figure 19. Schematic diagram of cold plate liquid cooling system for data centers (under the Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License) [45].

Additively manufactured pure copper cold plates via topology optimization break the design limitations of conventional regular rectangular and broken-line flow channels [46]. Flow channel density is adaptively arranged based on numerical simulation results of full-chip thermal load distribution: dense channels are laid out in zones corresponding to high-heat-flux dies, while simplified channel structures are adopted for low-heat-load regions. The microscale physical structure is displayed in Figure 20.

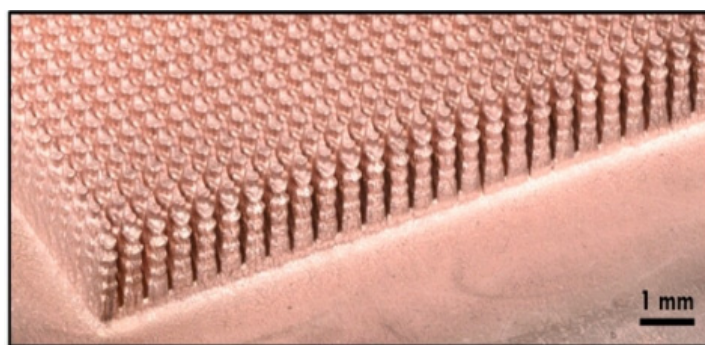


Figure 20. Microstructure close-up of topology-optimized pure copper cold plate fabricated via electrochemical 3D printing. Reprinted from Bazmi et al. (under the Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License) [46].

Compared with traditional pin-fin cold plates, this integrated electrochemical additive manufacturing solution reduces overall thermal resistance by 30% and flow pressure drop by 68%, delivering the best combined thermal and hydraulic performance among the four cold plate configurations. It can satisfy the stringent cooling specifications of high-end customized equipment with ultra-high computing power. Nevertheless, high-precision topological simulation and special metal additive manufacturing impose high technical barriers, resulting in extremely high unit production costs, which restricts its large-scale standardized deployment in data centers at the current stage.

A horizontal comparison of heat transfer capacity, flow loss, manufacturing cost and applicable scenarios across the four cold plate configurations is summarized as follows. Continuous serpentine cold plates feature simple processing and the lowest cost, targeting general servers with low-to-medium power consumption. Internally inclined rib-reinforced cold plates deliver moderate heat transfer performance and fit standard medium-power AI accelerator cards. Grid multi-split-flow cold plates exhibit remarkable temperature uniformity advantages, serving as the preferred cold plate solution for chiplet multi-die packaged chips. Topology-optimized additive-manufactured cold plates possess a good comprehensive thermal–hydraulic performance and are only adopted in high-end customized supercomputing equipment.

Two major technical lines can guide the subsequent optimization of cold plate liquid cooling: First, iterative development of bionic adaptive topological flow channel simulation algorithms to reduce overall flow pressure drop while maintaining cooling performance; second, improvements to low-cost metal additive forming processes to cut the manufacturing cost of complex split-flow cold plates, thereby promoting the large-scale commercial application of high-performance grid and topology-optimized cold plates.

3.1.4. Immersion Liquid Cooling

Immersion liquid cooling submerges all electronic components entirely in dielectric coolant. It realizes full-domain heat dissipation via single-phase convection or boiling phase change of working fluid. This technology eliminates numerous thermal interfaces and complex external pipelines, and features excellent temperature uniformity across the whole device and controllable PUE of computer rooms. It has become an advanced mainstream cooling technology for supercomputers and high-density AI cabinets. Its development follows an evolutionary route: basic single-phase mineral oil solution, two-phase channel enhancement with fluorinated fluid, overall cabinet layout optimization, cavity structure optimization, and passive immersion combined with heat pipes. Each generation of research continuously improves the deficiencies of previous working fluids, heat dissipation limits and cabinet layout drawbacks.

Schmidt et al. [47] established a single-phase immersion system using mineral oil, demonstrating the feasibility of whole-machine heat dissipation with a low-cost dielectric fluid and establishing the foundational engineering framework for immersion liquid cooling. Their work was instrumental in proving that direct liquid contact could effectively cool server components, but the mineral oil they employed exhibited inherently high viscosity and a relatively low heat transfer coefficient. As a result, this approach proved inadequate for kilowatt-level high-heat-flux chips and fell short of the thermal demands posed by emerging high-end computing hardware. To overcome these limitations, researchers shifted their focus toward two-phase immersion cooling, which utilizes low-boiling-point fluorinated fluids that leverage latent heat of vaporization for far superior heat removal. Fu et al. [48] adopted HFE-7100 as the working fluid and fabricated serrated turbulence structures inside the channels. Figure 21 compares the dimensional structures of straight channels and two types of serrated configurations. The serrated structures enhance boiling heat transfer by disturbing fluid flow and promoting bubble detachment, and greatly raise the critical heat flux through channel reinforcement, effectively lifting the limit of two-phase heat dissipation. However, this study only focused on single-channel heat transfer, without exploring the overall cabinet layout.

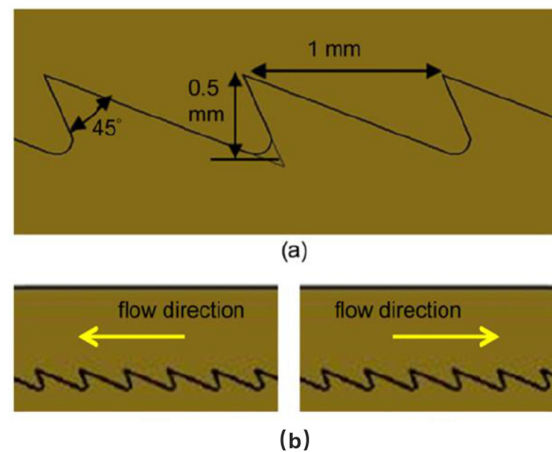


Figure 21. Schematic diagrams of (a) straight channel and (b) two types of serrated microchannels. Adapted from Fu et al. (under a Creative Commons Attribution 4.0 License) [48].

Liu and Yu [49] built a full-scale two-phase immersion test platform with Novec 7100 as the working fluid. Figure 22 shows the principle schematic of the complete immersion system, clearly illustrating the full circulation path of fluid boiling, vapor condensation and reflux. They mainly investigated the influence of server spacing on energy efficiency, and achieved a PUE as low as 1.037 by optimizing component arrangement, which solved the heat transfer deterioration caused by disordered component layout inside cabinets. Still, this research lacked refined optimization on inlet and outlet structures as well as heat dissipation baseplates of the tank. Muneeshwaran et al. [50] carried out single-phase immersion experiments with FC-40. By comparing T-type and Z-type inlet-outlet structures and various baseplate designs, they proved that the combination of T-type flow distribution and heat pipe baseplate could significantly reduce the system thermal resistance, filling the research gap in tank structure design. As single-phase FC-40 absorbs heat merely through sensible heat, it still suffers from poor temperature control under high-heat-flux conditions. To thoroughly break the boiling limit of conventional two-phase immersion and realize pump-free passive heat dissipation, Lu et al. [51] proposed a composite scheme combining immersion cooling with double-sided porous sintered flat heat pipes. Figure 23 displays the layered structure of the heat pipe. The inner and outer composite porous structures enhance boiling performance and internal vapor–liquid separation, respectively. Relying on the temperature uniformity of heat pipes and surface enhanced boiling, this scheme achieves passive heat dissipation of 1200 W for a single chip, far exceeding the heat dissipation limit of traditional immersion solutions.

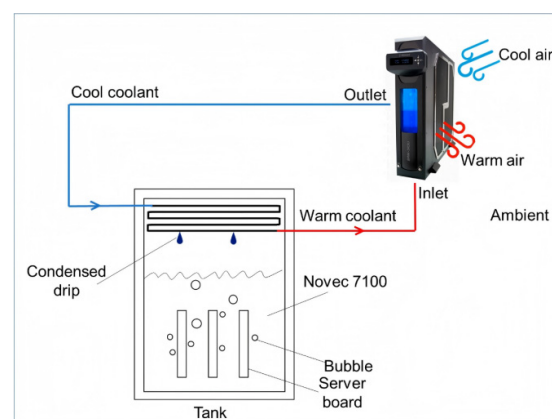


Figure 22. Schematic diagram of the working principle of the two-phase immersion liquid cooling system for data centers (under a Creative Commons Attribution 4.0 License) [49].

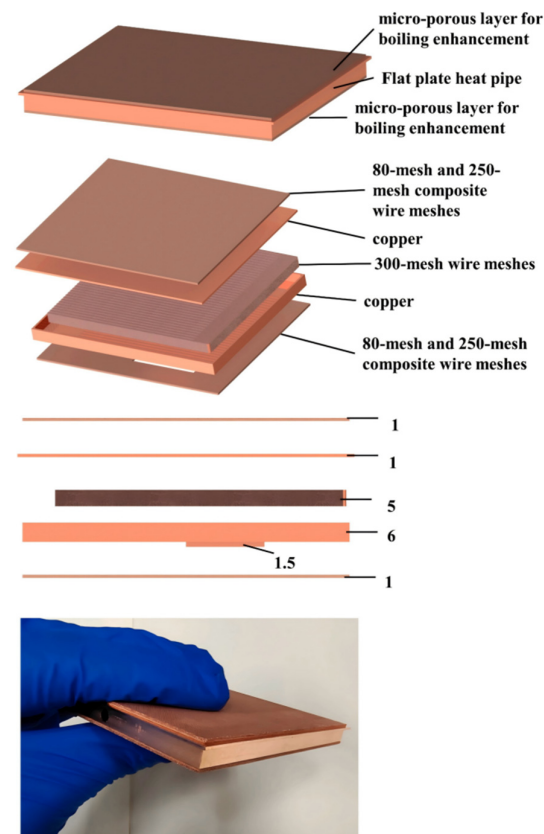


Figure 23. Schematic Diagram of structure and morphology of double-layer porous composite flat heat pipe (under a Creative Commons Attribution 4.0 License) [50].

For multi-chip modules, the thermal performance of two-phase immersion cooling has been systematically evaluated through numerical modeling [52]. The results indicate that coolant type, inlet velocity, and subcooling significantly affect chip temperatures, with Novec 7100 outperforming FC-72 due to its higher latent heat. A critical finding is that the incorporation of a baffle structure can reduce the temperature uniformity coefficient of the multi-chip module by up to 63.08%, substantially mitigating inter-chip temperature disparities [52]. This provides a practical pathway for improving thermal uniformity in densely packed processor clusters through flow-field optimization.

The technical evolution rules are summarized as follows. Single-phase mineral oil immersion is low cost and is suitable for renovating old server rooms with medium and low power density. Serrated-enhanced two-phase fluorinated fluid cooling applies to heat dissipation of single chips with low and medium power. The two-phase Novec system with optimized layout is suitable for large-scale standard AI cabinets. The single-phase FC-40 solution facilitates customized modification of tank structures. The passive immersion integrated with heat pipes delivers the best comprehensive performance, targeting ultra-high-power flagship chips and high-density supercomputer clusters. The overall technical improvement proceeds step by step, covering working fluid optimization, channel enhancement, cabinet layout, cavity structure and built-in thermal equalization components. Corresponding immersion solutions can be selected flexibly according to cabinet power density and construction cost.

3.2. Chip Package-Level Thermal Management

Chip package-level thermal management serves as the critical link between the chip and the system-level heat dissipation. It is responsible for rapidly extracting internal heat flux and mitigating thermal spikes and thermal fatigue. As power density continues to rise,

this field has evolved into a technology framework that integrates thermal interface materials, passive two-phase heat transfer devices, and phase-change materials (PCMs). Together, these approaches form the core enabler for the reliable operation of high-power chips.

3.2.1. Thermal Interface Materials

In chip package-level thermal management, TIMs serve as the key medium that fills micron-scale gaps between the chip and the heat sink to reduce contact thermal resistance. Traditional TIMs are mostly polymer-based materials such as thermal greases and thermal pads, which are commonly used for heat dissipation of low-to-medium power chips. However, once the chip power density exceeds 300 W/cm^2 [53], these conventional materials can no longer meet the heat dissipation requirements. Novel TIMs, such as liquid metals and diamond-composite materials, have become the research focus for heat dissipation in high-power packaging.

To understand why conventional polymer-based TIMs fail at high power densities, it is essential to examine the heat transfer mechanism at solid–solid interfaces. At the microscopic level, the temperature drop across a contact interface is governed by the interfacial thermal resistance, often referred to as the Kapitza resistance in the context of phonon-mediated transport:

$$\Delta T_{\text{int}} = R_{\text{int}} \cdot q \quad (6)$$

or, equivalently, through the thermal boundary conductance G :

$$q = G \cdot \Delta T_{\text{int}} \quad (7)$$

where R_{int} is the interfacial thermal resistance ($\text{m}^2 \cdot \text{K/W}$) and G is the thermal boundary conductance ($\text{W}/(\text{m}^2 \cdot \text{K})$). This resistance originates from acoustic/phonon mismatch, surface roughness, and incomplete physical contact at the interface. It explains a critical paradox: even if a TIM's bulk material has ultra-high thermal conductivity (e.g., diamond $> 1000 \text{ W}/(\text{m} \cdot \text{K})$), the overall thermal performance can still be severely degraded if the interface itself has low G . This is precisely why the subsequent developments—colloidal liquid metals (addressing wettability to enhance G), gradient-modified diamond composites (reducing phonon scattering at interfaces), and hybrid structural TIMs (creating multiple contact paths)—all focus on maximizing G through interfacial engineering rather than solely pursuing higher bulk thermal conductivity.

However, the application of liquid metals is constrained by their susceptibility to leakage and poor compatibility with substrate materials. Even though liquid metals exhibit superior wettability and lower interfacial thermal resistance compared with conventional polymer-based TIMs, their large-scale deployment has remained challenging. Consequently, the research community has been focusing on improvement strategies such as colloidal curing and interfacial modification. To address these practical issues, Wu et al. [54] pioneered a mechanochemistry-mediated approach to prepare colloidal liquid metals. By exploiting a liquid–solid interlocking interfacial structure, this method simultaneously achieves high heat flux dissipation and leakage prevention, thereby overcoming the key engineering limitations of liquid metals and significantly narrowing the gap from laboratory research to industrial application.

While the liquid metal route continues to improve, the high interfacial thermal resistance at the matrix interface has long constrained the performance of diamond-composite TIMs, preventing the ultra-high intrinsic thermal conductivity of diamond from being fully utilized. To overcome this interfacial impedance bottleneck, Zhang and his team [55] proposed a gradient interfacial modification strategy, establishing a full-chain design framework ranging from interfacial wettability regulation to thermal conductivity optimization.

Figure 24 systematically illustrates the development principles of metal/diamond composites. Panels (a) and (b) show the matching between thermal conductivity and coefficient of thermal expansion under different compositions, demonstrating the potential of diamond. Panels (c) and (d) reveal the underlying mechanism by which the gradient carbide transition layer weakens phonon scattering and opens up efficient heat transport pathways.

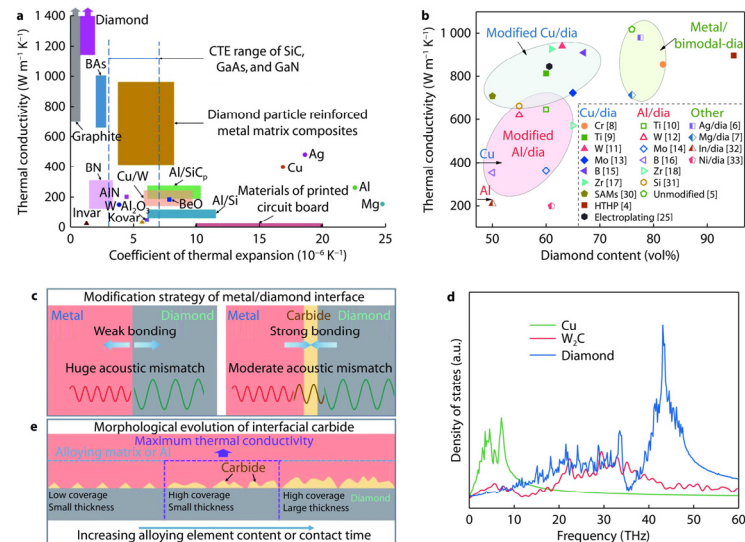


Figure 24. Schematic illustration of the thermal conductivity, interfacial modification strategies, and heat transfer mechanism of metal/diamond composites. Adapted from Zhang et al., 2023 [55]. (a) Comparison of thermal conductivity and coefficient of thermal expansion for typical semiconductor, substrate and metal/diamond composite materials; (b) Thermal conductivity distribution of Cu/diamond and Al/diamond composites with varying diamond volume fractions, including modified and bimodal-diamond reinforced systems; (c) Schematic diagram of acoustic mismatch at the metal-diamond interface before and after carbide interfacial modification; (d) Density of states curves of Cu, W₂C and diamond covering the terahertz frequency range; (e) Morphological evolution law of interfacial carbide layers with the increase of alloying element content or interfacial contact time, and its corresponding thermal conductivity performance. (under a Creative Commons Attribution 4.0 License).

In addition to these matrix and interface-modification approaches, a morphology-controllable liquid-metal/diamond sandwich structure has been developed, achieving an ultralow interfacial thermal resistance and improved extrusion resistance by tailoring the spatial arrangement of diamond particles within the liquid metal [56]. Meanwhile, Du et al. [57] systematically studied the microstructure evolution of diamond/GaSn composites under long-term thermal cycling, revealing that interfacial reaction-layer growth and porosity changes are primary causes of thermal conductivity degradation, a critical consideration for the long-term reliability of TIMs in high-power packaging.

While the two material improvement strategies described above have achieved performance breakthroughs through matrix modification and filler interface optimization, respectively, they still struggle to simultaneously realize ultra-high thermal conductivity and compatibility with mass production. Focusing on the pain points of industrial deployment, Lofgreen et al. [58] moved beyond pure material modification and pioneered a hybrid TIM architecture. This design uses metal thermal posts embedded in a polymer matrix to create primary and secondary cooperative heat transport paths, and was patented. Figure 25 shows the package cross-section of this hybrid TIM. Densely arranged metal thermal posts inside the structure form high-speed primary heat conduction channels, while the surrounding flexible polymer matrix fills the gaps and serves as the secondary

phase. This configuration achieves a thermal conduction level comparable to that of liquid metals through the metal pathways, while retaining the assembly and packaging process compatibility of conventional thermal greases thanks to the flexible matrix. This design addresses the engineering shortcomings of the previous two approaches and provides a mature, deployable solution for thermal management of high-power CPU packaging.

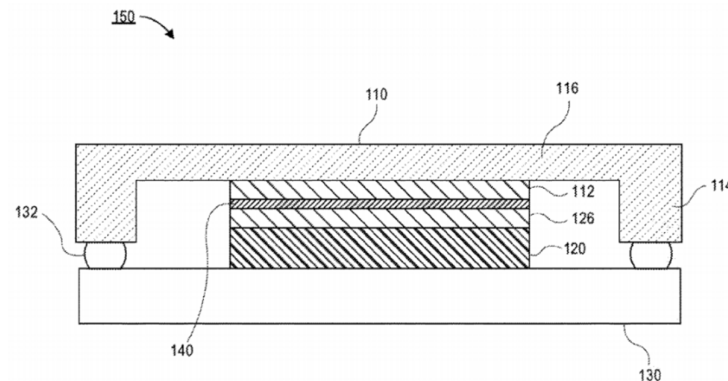


Figure 25. Packaging structure of the hybrid liquid metal TIM. Adapted from Lofgreen et al. (US Patent 11,551,994 B2) [58].

These three approaches form a progressive chain that addresses the shortcomings of their predecessors. To overcome the limited heat conduction of conventional polymers, research first turned to matrix modification to develop high-thermal-conductivity liquid metals, and solved the leakage problem through colloidal curing. Subsequently, interfacial modification was employed to optimize diamond composite fillers, breaking the bottleneck of interfacial thermal resistance. Finally, moving beyond the paradigm of pure material improvement, architecture integration has driven thermal interface materials toward system-level development that simultaneously achieves high thermal conductivity and high reliability.

3.2.2. Passive Two-Phase Heat Transfer Devices: Heat Pipes and Vapor Chambers

Heat pipes deliver efficient and rapid heat transfer via phase-change cycles of internal working fluid evaporation and condensation. Their equivalent thermal conductivity far exceeds that of solid metals, making them the most widely adopted passive two-phase heat dissipation components to date. Driven by capillary force generated from wicks, heat pipes circulate liquid working fluid without external pumps for fluid transportation, featuring prominent merits including compact structure, passive self-sustained operation and high reliability. Since their invention, multiple configurations such as micro heat pipes and loop heat pipes have been developed. Their application scope has expanded from aerospace thermal control to various thermal management systems for consumer electronics and power chips.

From the perspective of fundamental heat transfer mechanisms and mathematical modeling frameworks, the core driving force of the full two-phase cycle in heat pipes is capillary lift generated by menisci at solid–liquid interfaces within wicks. The complete cycle consists of four coupled heat and mass transfer processes: liquid film boiling and phase change in the evaporator, vapor diffusion and transportation inside the vapor chamber, film condensation and heat release in the condenser, and liquid backflow through capillary channels. Thermal resistance and flow resistance interact and restrict each other throughout all stages, jointly determining the critical heat transfer limit of the device. Classical fundamental theories define four core performance constraint thresholds for heat pipes: saturated temperature gradient limit, capillary limit, boiling limit and entrainment limit. Quantitative verifications prove that their phase-change heat transport capacity

greatly outperforms pure solid metal conduction, laying fundamental theoretical support for structural design, working fluid selection and operating condition matching of all heat pipe configurations [59].

To adapt to ultra-thin laptops, miniaturized chip packages and other highly compact devices and fill the heat dissipation gap of lightweight miniature electronics, targeted research [60] develops ultra-thin flat heat pipes. Abandoning traditional wick structures designed for thick chambers, researchers re-match vapor–liquid two-phase flow rules under minimal thickness and successfully fabricate ultra-thin flat heat pipes thinner than 1 mm. This achieves breakthroughs in heat pipe miniaturization and promotes large-scale application of heat pipes in consumer electronics. To meet unified design demands for various heat pipe types, existing comprehensive theoretical frameworks integrate phase-change mechanisms, flow resistance models and thermal resistance networks of conventional circular heat pipes, micro flat heat pipes, loop heat pipes and pulsating heat pipes. These frameworks systematically sort out coupled calculation methods for capillary force, vapor resistance and interfacial heat transfer coefficients across different configurations, establishing a universal mathematical analysis system covering all heat pipe categories. It provides complete theoretical tools for parametric optimization of heat dissipation schemes for ultra-high heat flux chips and new energy power devices [61].

Benefiting from the one-dimensional linear directional heat transfer advantage of heat pipes, long-distance heat transport challenges can be effectively resolved. Nevertheless, chips and power modules mostly feature planar concentrated heat sources. A single linear heat pipe only enables unidirectional heat conduction at discrete points and fails to equalize heat across the entire two-dimensional plane, easily inducing localized high-heat-flux hotspots over heat source regions. Vapor chambers serve as complementary two-dimensional planar two-phase heat spreaders to address such planar heat equalization demands. With flat hermetic cavity structures, vapor chambers are equipped with double-layer wicks on the evaporator and condenser sides of upper and lower plates, while the middle hollow space acts as vapor diffusion channels. Heat can spread rapidly in all two-dimensional planar directions, fundamentally mitigating hotspot effects from concentrated heat sources and becoming core heat equalization solutions for highly integrated electronic devices.

Early heat transfer simulations and structural optimizations of vapor chambers adopt uniform heat flux assumptions for the entire domain, which deviate significantly from actual service conditions of chips with localized concentrated high heat flux and partitioned variable power. Uniform heat source models cannot accurately predict liquid film dry-out evolution in local hotspot zones, failing to support refined engineering design of cavity thickness, graded capillary porosity and support array layout. To resolve mismatches between theoretical assumptions and practical operating conditions, existing non-uniform heat transfer models introduce partitioned variable heat flux boundary conditions. They quantitatively analyze the full evolution of liquid film thinning, dry spot expansion and two-phase interface instability degradation induced by localized high thermal loads, and distinguish differentiated calculation coefficients for local boiling heat transfer in evaporators and film condensation heat transfer in condensers. A full-domain thermal resistance calculation framework coupled with capillary backflow resistance and lateral vapor diffusion resistance is established, breaking the applicable boundary of uniform heat source theories and delivering precise theoretical support for graded wick design, critical heat flux prediction and cycle life evaluation of vapor chambers [62].

Microelectronic devices keep evolving toward ultra-thin, miniaturized and highly integrated packaging, continuously compressing the overall cavity thickness of vapor chambers. The cross-sectional area of vapor flow channels and the thickness of wick pores

shrink simultaneously, triggering two major scale-effect drawbacks. First, reduced wick thickness narrows liquid flow cross-sections and drastically lowers capillary backflow permeability, readily inducing capillary-limit dry spots under identical thermal loads. Second, narrow vapor channels inside ultra-thin chambers lead to exponentially rising lateral vapor diffusion resistance, generating vapor locks that block liquid replenishment and exacerbate local overheating failures. To tackle two-phase heat transfer bottlenecks under ultra-thin dimensions, existing review papers systematically summarize a full set of optimization technical routes for ultra-thin vapor chambers, covering four core optimization directions: composite wicks with graded pore sizes, drag-reduction structures with micro-pillar supports, integrated cavities fabricated via metal additive manufacturing, and composite modified working fluids with low surface tension. They clarify the primary and secondary restrictive relationships among capillary resistance, vapor resistance and interfacial thermal resistance under ultra-thin scales [63].

Figure 26 fully illustrates the internal structure and self-sustained phase-change cycle mechanism of hermetic ultra-thin vapor chamber cavities. It marks partitioned regions including evaporative boiling heat transfer zones, vapor diffusion channels, condensing film heat release zones and multi-layer capillary backflow channels. The complete coupled processes—liquid full-plane backflow driven by capillary pressure difference and two-dimensional lateral vapor heat spreading—are clearly demonstrated. This provides a complete theoretical design basis for lightweight and highly stable structural optimization of vapor chambers deployed in next-generation ultra-high-power ultra-thin packaged electronic devices.

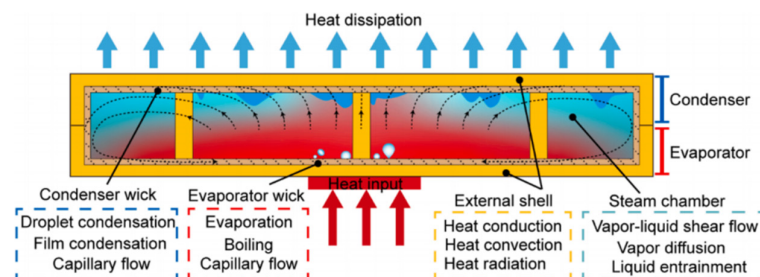


Figure 26. Schematic diagram of the structure and working principle of a vapor chamber (reproduced with permission) [63].

By analyzing the technological evolution of the two types of passive two-phase heat spreaders, the industry has developed a hierarchical technical roadmap to meet the rising heat flux demands of chips: one-dimensional directional heat transport via heat pipes and two-dimensional planar heat spreading via vapor chambers. Heat pipes rely on self-sustained capillary phase-change cycles as their core mechanism. Separated loop topologies break the constraints of gravity and long-distance flow resistance, while unified mathematical models for all heat pipe configurations refine the prediction framework of critical heat transfer limits. Such devices fully address the demand for efficient long-distance heat transport, yet their one-dimensional linear heat transfer structure cannot eliminate localized hotspots on planar heat sources. Vapor chambers adopt flat cavity-based two-dimensional heat spreading architectures to compensate for the poor planar temperature uniformity of heat pipes. Researchers have successively overcome theoretical limitations from uniform heat flux assumptions and resolved two-phase flow resistance failures under ultra-thin dimensions, making vapor chambers suitable for high-density miniaturized chip packaging. Overall, heat pipes and vapor chambers complement each other. Together they form a comprehensive multi-dimensional technical framework for passive phase-change cooling, outlining lightweight, pumpless, and long-lifetime reliable

optimization routes for thermal management of ultra-high-computing chips and new-energy power devices.

3.2.3. Phase-Change Materials (PCMs)

In high-power chip packaging, severe fluctuations in transient thermal loads often cause sudden rises and drops in junction temperature, leading to thermal fatigue failure. Phase-change materials, by virtue of their ability to absorb and release large amounts of latent heat during solid–liquid phase transition, have become an ideal choice for package-level thermal buffer layers.

Li et al. [64] conducted a specialized study on three-dimensional integrated circuit packaging. As shown in Figure 27, they incorporated expanded graphite into a PCM composite and integrated it into the package structure. Under experimental conditions, they achieved a constant temperature regulation of $\pm 5\text{ }^{\circ}\text{C}$, demonstrating the application potential of PCMs in chip thermal management. They also identified the urgent challenges associated with conventional PCMs, such as poor thermal conductivity and susceptibility to leakage. Building on this foundational research, further studies moved into the engineering phase. Two studies, each with its own merits and complementing each other, are worth noting. Hollis et al. [65] introduced PCMs into a silicon carbide power electronic packaging system in Figure 28. Using simulation tests under electric vehicle dynamic thermal loads, they reduced the chip junction temperature fluctuation amplitude by more than 30%. From the perspective of application scenarios, the work by Li et al. is fundamental in nature, suitable only for laboratory environments and not directly guiding industrial applications. The work by Hollis et al., while focusing on engineering implementation, did not address the issues of thermal conductivity and leakage highlighted earlier; the inherent drawbacks of the material remain. Nevertheless, this achievement completed the engineering validation of the technology based on Li's foundation. Together, the two form a complete research progression, successfully advancing PCM thermal management technology from the stage of fundamental laboratory theory to practical engineering implementation in power electronic devices. This provides solid theoretical support and experimental evidence for the industrial application of phase-change encapsulation temperature control technology, significantly accelerating the transition of PCM technology from theoretical research to engineering practice.

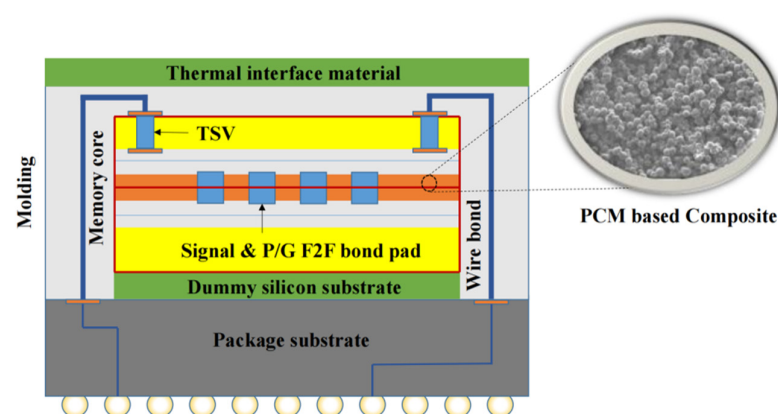


Figure 27. Schematic diagram of the integration scheme of PCM-based composite materials in 3D IC packaging under the Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License [64].

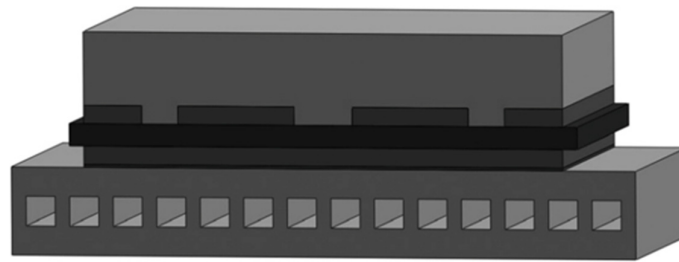


Figure 28. Schematic diagram of the single-sided packaging structure of a SiC power device integrated with PCM (reproduced with permission) [65].

To overcome the common drawbacks of conventional organic PCMs, such as large supercooling degree and slow response speed, in 2024 Ki et al. [66] developed a porous copper-composited gallium-based liquid metal PCM, as shown in Figure 29. Using an interfacial modification strategy, this approach reduced the material's supercooling degree by 63%. In transient tests on a microheater, the temperature fluctuation suppression ratio reached 21.3%, with a peak temperature reduction of 9.2 °C, effectively compensating for the performance shortcomings of conventional organic PCMs and further broadening the application range of PCMs in room-temperature chip packaging. Unlike experimental research focused on material performance improvement, Zheng et al. [67] adopted a global industry perspective and comprehensively reviewed the development history of PCM thermal management technology for electronic devices in a related review article. They clearly pointed out that future efforts for package-level PCMs still need to focus on cyclic stability, packaging process compatibility, multi-functional integration, and other areas, thereby clarifying the development path and pointing out the direction for subsequent research in the entire field.

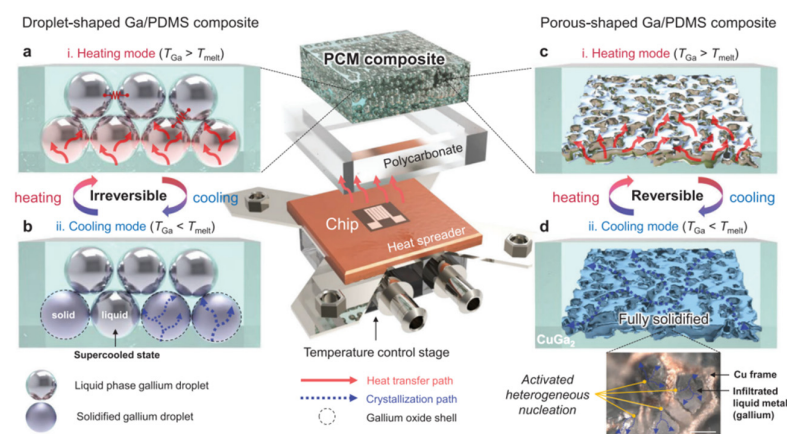


Figure 29. Comparison of thermal regulation mechanisms between droplet-type and porous-type gallium-based PCM. Adapted from Ki et al. [66] under the Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License. (a) Heating working mode of droplet-shaped Ga/PDMS composite when gallium temperature exceeds melting point, showing internal heat transfer paths inside liquid gallium droplets; (b) Cooling working mode of droplet-shaped Ga/PDMS composite when gallium temperature drops below melting point, presenting the supercooled state and irreversible crystallization paths of gallium droplets wrapped by gallium oxide shells; (c) Heating working mode of porous-shaped Ga/PDMS composite under high temperature, illustrating uniform heat circulation channels inside the interconnected porous liquid metal network; (d) Cooling working mode of porous-shaped Ga/PDMS composite at low temperature, revealing full solidification of gallium to form CuGa₂ alloy and activated heterogeneous nucleation effect inside the copper frame skeleton, which enables reversible heating-cooling cycle performance.

In the most recent study achieving a breakthrough in material performance, Wu et al. [68] proposed a composite PCM system featuring a directional thermal conduction network, breaking through the traditional limitation that high thermal storage and high thermal conductivity cannot be simultaneously achieved. The thermal conductivity of this material is more than ten times higher than that of pure paraffin. When applied to high-power chips, it reduces the peak temperature under transient conditions by more than 15 °C, overcoming the common drawback of conventional PCMs—fast heat storage but slow heat release. This provides a new approach for the optimal design of PCMs for high-performance chip packaging.

3.2.4. High-Thermal-Conductivity Materials

High-thermal-conductivity material integration technology uses diamond, silicon carbide (SiC), two-dimensional materials, and other high-thermal-conductivity components to modify traditional silicon-based structures. It breaks through the intrinsic thermal conductivity limit of silicon. It builds low-resistance and efficient heat conduction paths. These paths quickly extract heat from the chip core. This improves device thermal stability and operational reliability.

This technology system includes two mainstream near-junction heat dissipation paths: heterogeneous integration of high-thermal-conductivity substrates, and on-chip heat conduction pathways. These paths suit different integration architectures and heat flux conditions.

Heterogeneous integration of high-thermal-conductivity substrates optimizes heat dissipation from the heat source base layer. It replaces traditional silicon substrates with SiC or diamond, which have much higher thermal conductivity than silicon. Using a low-temperature bonding process, it achieves a dense bond between the chip and the substrate. This reduces the device junction temperature at the source.

For example, gallium nitride (GaN)/aluminum gallium nitride (AlGaN) high electron mobility transistors (HEMTs) on SiC substrates [69] benefit from SiC's much higher thermal conductivity than silicon. SiC meets both RF insulation and vertical heat transfer requirements. Under the same power conditions, it significantly lowers the device junction temperature and improves radio frequency (RF) power density. This shows potential for industrial applications.

Regarding the problem that high-temperature processes damage devices when using diamond composite substrates, existing low-temperature integration schemes reconstruct the GaN–diamond interface [70]. They achieve direct bonding of nanocrystalline diamond with GaN, avoiding irreversible performance damage from high temperatures. As shown in Figure 30, the integration of nanocrystalline diamond with GaN not only effectively avoids the irreversible damage caused by traditional high-temperature processes but also greatly reduces the interfacial thermal resistance, further optimizing heat dissipation performance.

To precisely quantify the variation of interfacial thermal resistance, Cho et al. [71] conducted systematic thermal characterization on GaN–diamond composite substrates. They clarified the regulatory effect of the transition layer structure on overall heat dissipation. They refined the heat dissipation mechanism of high-thermal-conductivity substrates. This provides experimental and theoretical support for interfacial structure optimization.

In addition to substrate replacement, on-chip heat spreading and thermal conduction pathway technologies introduce high-thermal-conductivity media such as copper thermal through-silicon vias (TTSVs) and graphene without changing the silicon substrate itself. They construct a vertical-and-in-plane integrated thermal conduction network. This achieves efficient hotspot heat dissipation and temperature equalization. These schemes are compatible with CMOS processes and have relatively low engineering costs.

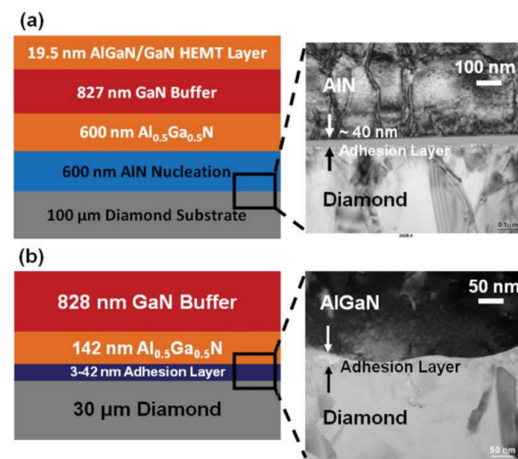


Figure 30. Schematic cross-sectional structures of two generations of GaN–diamond composite substrates (reproduced with permission) [70]. (a) Layer structure diagram of the first-generation GaN-on-diamond substrate with a 100 µm thick diamond base, including a 600 nm AlN nucleation layer, 600 nm $\text{Al}_{0.5}\text{Ga}_{0.5}\text{N}$ transition layer, 827 nm GaN buffer and 19.5 nm AlGaIn/GaN HEMT functional layer; the corresponding TEM image shows a ~40 nm-thick AlN adhesion layer at the diamond interface with a 100 nm scale bar. (b) Layer structure diagram of the optimized second-generation GaN–diamond composite substrate adopting a thinner 30 µm diamond substrate, which cancels the thick AlN nucleation layer and replaces it with a 3–42 nm ultra-thin adhesion layer, accompanied by a reduced 142 nm $\text{Al}_{0.5}\text{Ga}_{0.5}\text{N}$ interlayer and 828 nm GaN buffer; the matched TEM micrograph characterizes the ultra-thin adhesion layer region with a 50 nm scale bar.

TTSV arrays serve the function of enhancing vertical heat transfer in stacked chips [72]. The simulation of the stacked structure in Figure 31 visually verifies their effect in reducing chip peak temperature. Via dimensions and array arrangement directly change the total system thermal resistance. Parametric analysis can establish a quantitative relationship between structural dimensions and heat dissipation performance.

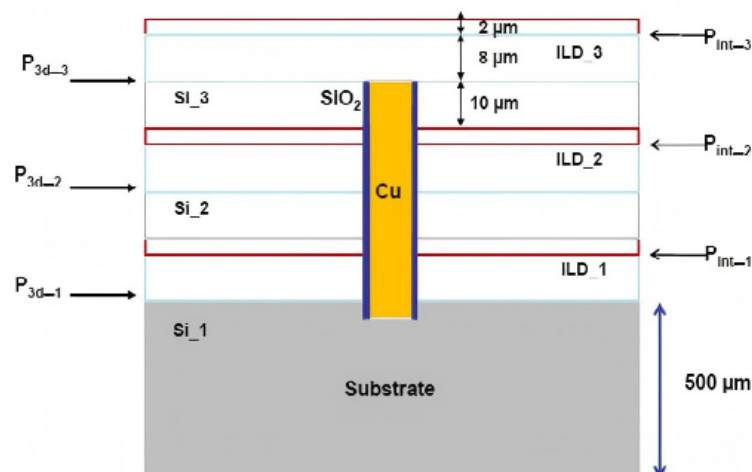


Figure 31. Schematic diagram of a 3D integrated circuit stacked structure with integrated thermal through-silicon vias (TTSVs) (reproduced with permission) [72].

Graphene can serve as a lateral heat spreading layer to complement heat dissipation [73]. With its outstanding in-plane thermal conductivity, it disperses locally concentrated heat and suppresses hotspot temperature rise. This enriches the application theory of two-dimensional materials in on-chip thermal management scenarios.

After the thermal benefits of both vertical and lateral on-chip heat conduction structures have been verified, a thermal resistance matrix model for multi-layer stacked inte-

grated circuits has been constructed. This model equates the silicon substrate, thermal conduction media, and interlayer interfaces within the chip as standardized thermal resistance units [74]. It quantitatively breaks down the heat transfer contributions of each multi-level thermal conduction path. It fully explains the underlying regulatory mechanisms of the internal thermal conduction network on the overall chip temperature distribution and hotspot magnitude. This provides a reliable theoretical tool for rapid temperature field prediction in complex three-dimensional stacked structures.

Comparing the two types of high-thermal-conductivity cooling solutions, heterogeneous substrate integration offers a higher heat dissipation limit and is more suitable for high-power wide-bandgap devices. However, it faces challenges such as substrate replacement, complex bonding processes, and relatively high manufacturing costs. On-chip embedded thermal conduction pathways do not require substrate reconstruction, offering clear advantages in process compatibility and mass production. They are better suited for traditional silicon-based 3D stacked chips. Their only drawback is that under ultra-high heat flux scenarios, their heat dissipation upper limit is lower than that of substrate replacement solutions. In practical engineering, the two approaches can be used in combination based on chip power levels and manufacturing conditions to achieve comprehensive thermal optimization.

3.2.5. Thermal Co-Design and Advanced Packaging for 3D IC and Chiplet

3D IC stacking and chiplet heterogeneous packaging are key architectures in the post-Moore era. They break the limit of single-chip computing power. They also integrate HBM high-bandwidth memory. They stack multiple dies vertically. They package many functional chiplets together. This greatly increases computing density.

However, these architectures have three inherent thermal drawbacks. First, multi-layer die stacking creates high vertical thermal resistance. Second, there is strong thermal coupling between layers. Third, lateral thermal crosstalk occurs across chiplets. In addition, computing chiplets and HBM dies are placed very close to each other. This generates local ultra-high heat flux.

The traditional sequential design approach (circuit, then package, then cooling) cannot accurately predict full-chip temperature gradients, hotspot magnitudes, or long-term thermal cycling reliability. Therefore, we need an integrated co-design system that combines architecture, packaging, and cooling. This system performs multi-physics cross-layer optimization during front-end thermal-aware floor planning. It simultaneously reduces vertical thermal resistance, interlayer coupling, and lateral crosstalk. It does so by optimizing chiplet placement, interconnect media, interlayer cooling, and memory layout. This achieves the best global balance among performance, power loss, and thermal reliability.

Vertical stacked architectures suffer from inherent fundamental heat transfer limitations, including intrinsically high vertical thermal resistance and severe interlayer temperature gradients. To address these drawbacks, a synergistic coupling scheme integrating microfluidic interlayer cooling and TSV arrays has emerged as a fundamental approach to enhance interlayer heat transfer. Within this architecture, microscale liquid cooling channels are embedded in the interlayer gap between two stacked dies, while high-density metallic thermal TSV arrays are deployed to form composite vertical heat conduction paths. Penetrating the interposer and underfill dielectric, the TSV arrays directly connect the active regions of the upper and lower dies, drastically reducing the equivalent vertical thermal resistance induced by series-connected multi-layer dielectrics. Meanwhile, the interlayer microfluidic channels continuously dissipate accumulated interlayer heat, mitigating vertical thermal gradients bidirectionally.

This scheme features a complete set of tunable quantitative design parameters. The areal density of TSV arrays can be modulated within the range of 500–5000 vias per square millimeter, with the diameter of individual TSVs and the center-to-center pitch between adjacent TSVs directly governing the coverage ratio of vertical heat conduction paths. Three types of substrates are available for interposers: silicon interposers (thermal conductivity: $148 \text{ W}/(\text{m}\cdot\text{K})$), organic interposers ($0.2\text{--}0.8 \text{ W}/(\text{m}\cdot\text{K})$), and glass interposers ($1.0\text{--}1.5 \text{ W}/(\text{m}\cdot\text{K})$), whose thermal conductivities differ by up to two orders of magnitude. Conventional underfill adhesives for stacking gaps exhibit low thermal conductivity of merely $0.3\text{--}0.6 \text{ W}/(\text{m}\cdot\text{K})$; upon modification with high-thermal-conductivity composite fillers, this value can be elevated to $1.2\text{--}2.5 \text{ W}/(\text{m}\cdot\text{K})$, substantially alleviating the thermal insulation effect of interlayer dielectrics.

Quantitative simulation results for the coupled model have been validated in prior research [75]; the quantitative correlation between TSV density and interlayer thermal resistance is documented in system-level simulation work [76], and heat exchange performance data of stacked interlayer microchannels are reported in Ref. [77]. Under operating conditions with a chiplet stacking thickness of $50 \mu\text{m}$ and a single chiplet power consumption of 40 W , increasing the TSV density from $800 \text{ vias}/\text{mm}^2$ to $3200 \text{ vias}/\text{mm}^2$ and raising the underfill thermal conductivity from $0.4 \text{ W}/(\text{m}\cdot\text{K})$ to $2.0 \text{ W}/(\text{m}\cdot\text{K})$ reduces the total vertical interlayer thermal resistance by 47% and cuts the inherent temperature difference between upper and lower dies from 42 K to 16 K . This drastically suppresses interlayer thermal coupling originating from vertical stacking at the hardware structural level [78].

Nevertheless, this scheme only targets vertical heat transfer enhancement within stacked interlayers. It fails to regulate lateral thermal crosstalk arising from planar chiplet placement and HBM partition layout, rendering it incompatible with complex planar heat source distributions in heterogeneous co-packaging with multiple chiplets.

As chiplet heterogeneous integration continues to scale up, the number of chiplets in a single package increases to 8–16, and the planar center-to-center spacing between chiplets shrinks to $0.5\text{--}2 \text{ mm}$. Multiple HBM memory dies and high-performance computing chiplets are placed close together. This significantly amplifies lateral thermal crosstalk. At the same time, multi-layer stacking and multi-dielectric coupling introduce a huge number of geometric and material variables. Traditional detailed finite-element thermal simulations can take several hours for a single case, which cannot support fast iterative screening in front-end floor planning.

For hybrid packaging scenarios combining wide-bandgap (WBG) dies and AI chiplets, double-sided integrated liquid cooling serves as an effective approach to reduce thermal resistance within stacked layers. The research group led by Iannuzzo proposed a double-sided cooling packaging scheme for press-pack SiC devices. Heat exchange channels are synchronously arranged on both upper and lower substrates; when combined with high-thermal-conductivity filling media, the overall cross-layer thermal resistance of the device can be reduced by 42%, which greatly alleviates the severe temperature gradient generated by vertical stacking [79].

For fast electrothermal co-simulation of large-scale chiplet heterogeneous packaging systems, L. Codecasa et al. developed a series of ultra-fast compact thermal modeling tools oriented to integrated circuits, including the FANTASTIC and TRAC platforms, which break the computational efficiency bottleneck of traditional finite element thermal simulation [80]. Their parametric moment-matching compact thermal modeling method can automatically extract equivalent thermal resistance–capacitance networks from chip geometric and material parameters, and seamlessly couple with SPICE circuit simulation flows to realize fully coupled electrothermal prediction of multi-die stacked architectures. Compared with conventional finite element simulation which takes hours for a single work-

ing condition, Codecasa's model reduces single-case calculation time to the millisecond scale while maintaining prediction error below 5%, providing a lightweight numerical analysis foundation for early chiplet floor planning thermal screening. The tool chain fully supports multi-interposer, multi-TTSV and HBM co-packaged complex thermal domains, and can quantitatively characterize horizontal thermal crosstalk between adjacent chiplets and vertical interlayer thermal gradient evolution under dynamic power loads, filling the gap of fast high-precision thermal prediction tools for high-power AI chip packaging.

A compact thermal model based on regression algorithms [81] has become an efficient quantitative evaluation tool. This model takes as input quantitative variables such as chiplet spacing, TSV array density, interposer thermal conductivity, underfill thermal conductivity, and the lateral placement distance between HBM and computing chiplets. It can instantly output three key thermal metrics: global peak temperature, interlayer vertical thermal resistance, and the temperature rise due to lateral thermal crosstalk across chiplets. This eliminates the need for repetitive high-precision multi-physics simulations and supports rapid iteration in thermal-aware floor planning. For reference, the complete theory of the model iteration tool 3D-ICE 4.0 is given in [82].

Quantitative comparison data come from the UCLA heterogeneous chiplet simulation case [83]. When chiplet center-to-center spacing is reduced from 2 mm to 0.6 mm, the additional temperature rise caused by lateral thermal crosstalk between adjacent chiplets can reach 27 K. If HBM memory dies are placed close to one side of a high-power computing chiplet, the average temperature in the memory region increases by 18 K. However, using a partitioned isolated HBM layout can limit this additional temperature rise to within 6 K. Additional verification of HBM isolation layout temperature control can be found in [84].

The compact thermal model can quickly traverse hundreds of combinations of chiplet placement, HBM layout, and TSV density matching schemes. It can screen out thermally degraded floorplan structures in the early design stage, providing quantitative decision-making support for thermal-aware chiplet floor planning. However, this model only performs steady-state thermal field calculations and does not include electrothermal coupling effects in multi-layer stacking systems. Therefore, its thermal evaluation results have inherent deviations.

Relying solely on package cooling structure optimization and fast steady-state thermal simulation models still cannot reproduce the true steady-state temperature rise behavior of 3D/chiplet integrated devices. In multi-layer stacking structures, the dielectric leakage power, underfill, and interposer material electrical parameters have a strong positive coupling with temperature. As temperature rises, dielectric leakage current and additional heat generation increase simultaneously, forming a self-amplifying temperature rise loop.

Quantitative control test data [85], obtained through a multi-physics electrothermal coupling test platform, show that under a standard 3D stacked chiplet operating condition of 45 W, the difference in steady-state peak hotspot temperature between a full electrothermal coupled simulation, and a pure thermal model that ignores electrothermal coupling can reach 20.8 K. If this coupling mechanism is neglected during the thermal-aware floor planning stage (which includes chiplet spacing planning, HBM layout, and TSV density matching), the actual peak operating temperature of the device will be significantly underestimated. This can lead to reliability risks such as insufficient design margin for thermal interface material selection, aggravated interlayer thermal cycling stress, and long-term packaging delamination failures.

Figure 32 quantifies the difference between the two models through temperature contour maps and axial temperature curves. The simulation results that do not include electrothermal coupling not only significantly suppress the peak hotspot temperature but also produce calculation deviations of 10–25 K in the regional temperature gradients caused

by lateral thermal crosstalk across chiplets. This fully demonstrates that electrothermal coupling quantification must be incorporated into the full-process thermal co-design system for 3D ICs and chiplets.

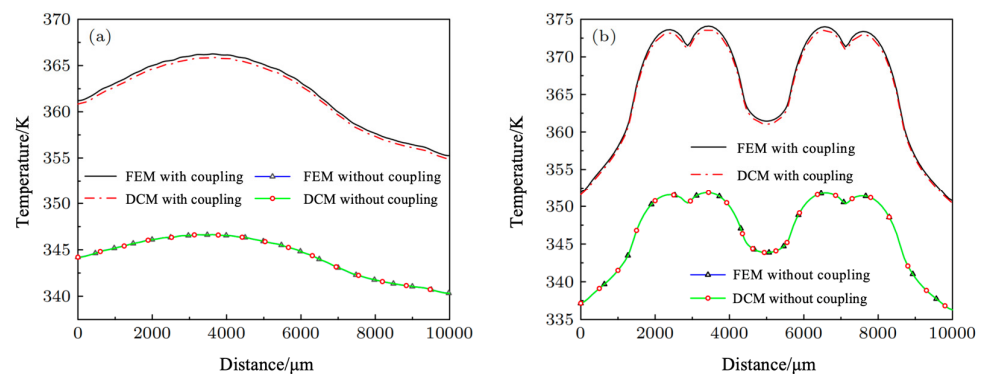


Figure 32. Chip temperature distribution without coupling and with coupling (reproduced with permission) [85]. (a) Lateral temperature profile of single heat source chip along the horizontal distance from 0 to 10,000 μm; four groups of curves are plotted to compare results of FEM with coupling, DCM with coupling, FEM without coupling and DCM without coupling. (b) Lateral temperature distribution curve of dual heat source chip within the distance range of 0–10,000 μm; the upper two curves represent coupled calculation results from FEM and DCM, while the lower two curves show uncoupled temperature data obtained by FEM and DCM methods respectively.

As technology evolves to the stage of large-scale chiplet heterogeneous advanced packaging, computing chiplets, multiple HBM memory chiplets, I/O interposers, and TSV vertical stacking layers are highly coupled. Multi-dimensional parameters—such as chiplet spacing, HBM parallel layout topology, interposer substrate thermal conductivity, underfill thermal conductivity, and TSV array density—mutually constrain each other. The traditional step-by-step design approach, which separates circuit architecture, packaging media, and interlayer cooling, cannot simultaneously balance multiple objectives including computing power, power consumption, global temperature uniformity, and packaging reliability.

To address this scenario, an integrated architecture-package-cooling thermal co-design method [86] has been proposed. This method establishes a closed-loop quantitative optimization process that connects front-end architecture definition, chiplet floor planning, packaging media selection, and interlayer microchannel cooling design. It incorporates all key quantitative design parameters required by reviewers into a unified optimization space: planar chiplet center-to-center spacing, HBM isolation placement distance from computing chiplets, interposer intrinsic thermal conductivity, TSV array areal density, underfill thermal conductivity in the stacking gap, thermal-aware global routing density, and microchannel geometry dimensions. Through multiple iterative cycles, it simultaneously regulates three core heat transfer metrics—vertical interlayer equivalent thermal resistance, additional temperature rise from lateral thermal crosstalk across chiplets, and interlayer coupled heat flux—to achieve a multi-objective Pareto optimal solution.

Figure 33 fully presents the closed-loop process of this cross-layer collaborative optimization flow. At the architecture layer and the advanced packaging layer, a bidirectional quantitative parameter feedback interface is established. Key design variables—including chiplet placement spacing, HBM partition layout, TSV array density, and thermal conductivity of the interposer and underfill—are used as core interactive variables. The closed-loop optimization sequentially performs design space parameter sweeping, multi-physics thermal metric quantitative evaluation, and iterative thermal-aware floorplan revision.

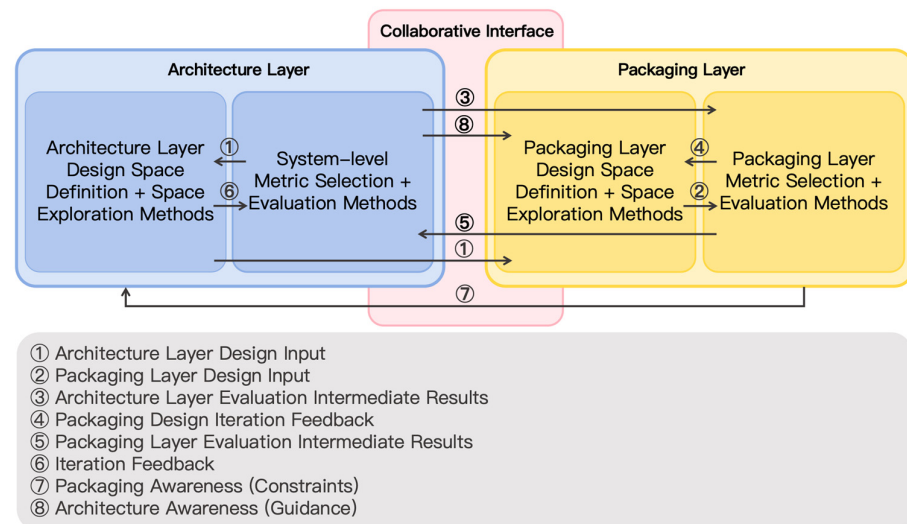


Figure 33. Schematic diagram of the architecture-package co-design methodology.

A quantitative engineering case comes from a collaborative optimization experiment on a 12-chiplet + 4-stack HBM co-packaged AI processor [35]. Using the co-design approach, the following configuration was adopted: chiplet spacing of 1.2 mm with partitioned placement, a high-thermal-conductivity silicon interposer combined with underfill of $1.8 \text{ W}/(\text{m}\cdot\text{K})$, a uniform TSV array of $2800 \text{ vias}/\text{mm}^2$, and an isolated layout between HBM and computing chiplets. As a result, the total vertical system thermal resistance was reduced by 52%, the additional temperature rise due to lateral thermal crosstalk across chiplets dropped from 29 K to 7 K, and the interlayer vertical temperature difference was controlled within 12 K. This systematically addresses the core pain points of excessive vertical thermal resistance, severe interlayer thermal coupling, and strong lateral thermal crosstalk in large-scale chiplet co-packaging, providing a complete cross-scale quantitative thermal management solution for next-generation high-power heterogeneous AI integrated chips.

Summarizing the complete optimization trajectory of thermal co-design technologies for 3D ICs and chiplets, the entire framework continuously improves quantitative design capabilities around three core heat transfer constraints: vertical thermal resistance, interlayer thermal coupling, and lateral thermal crosstalk. The evolution starts from TSV–microfluidic hybrid structures that regulate stacked vertical thermal resistance and quantitatively match TSV density with filler material thermal conductivity to alleviate interlayer temperature differences. It then moves to compact thermal models that enable rapid batch thermal evaluation of chiplet spacing, HBM layout, and interposer parameters. Next, it quantifies the 20.8 K temperature rise deviation caused by electrothermal coupling, refining the thermal design constraint boundaries. Finally, it culminates in a fully parameter-integrated cross-layer thermal co-design for large-scale chiplet architectures, achieving a transition from single-stack interlayer cooling structure optimization to system-level thermal design that covers chiplet floor planning, HBM partition layout, interposer media selection, TSV array quantitative matching, and thermal-aware placement.

Future thermal management innovations for 3D IC and chiplet integrated systems can focus on five major research directions: collaborative quantitative matching of multi-scale parameters; partitioned isolated HBM thermal-aware floor planning; non-uniform chiplet placement topologies with low lateral crosstalk; development of high-thermal-conductivity composite interposers and modified underfill materials; and precise control of vertical thermal resistance via gradient-density TSV arrays. These efforts will further weaken interlayer thermal coupling and lateral thermal crosstalk, meeting the extreme thermal

management demands of next-generation ultra-high computing power, multi-chiplet, HBM-co-packaged AI processors.

3.3. Comprehensive Comparative Analysis of Chip-Level Thermal Management Technologies

As power density and integration level of high-power integrated processors keep rising, a single cooling technology cannot simultaneously satisfy multi-dimensional requirements including ultra-high heat flux dissipation, temperature uniformity, manufacturing cost, process compatibility and long-term reliability. In recent years, various chip-level thermal management technologies have evolved into two major technical branches: near-junction liquid cooling and package-level thermal management. These two categories differ substantially in heat transfer mechanisms, installation positions and engineering characteristics, yet they feature strong complementarity. A comprehensive comparison covering cooling capacity, engineering maturity, manufacturing complexity and applicable scenarios is therefore necessary to guide thermal scheme selection for high-power integrated processors.

Near-junction liquid cooling achieves intensified heat exchange directly adjacent to heat sources and cuts overall thermal resistance by shortening heat conduction paths, delivering state-of-the-art cooling performance. Microchannel liquid cooling delivers coolant directly to chip near-junction regions. Continuous structural advances, including straight, finned, two-phase, embedded and manifold microchannels, steadily boost its heat dissipation capacity and temperature uniformity, supporting heat flux levels of hundreds of watts per cubic centimeter or above [23–26,28–34]. In contrast, direct jet impingement cooling disrupts thermal boundary layers via high-speed jets and excels at suppressing localized hotspots, making it ideal for processors with highly concentrated heat generation. Nevertheless, its cooling performance is highly sensitive to nozzle layout and flow field design [38–40]. Cold plate liquid cooling offers slightly weaker heat exchange capacity than direct near-junction liquid cooling, yet it boasts mature manufacturing, high reliability and convenient system integration, rendering it the most widely deployed liquid cooling solution for servers and high-performance computing platforms [41–45]. Immersion liquid cooling dissipates heat for entire electronic systems through dielectric coolants, providing outstanding temperature uniformity and energy efficiency for ultra-high-power computing hardware and data centers. Its large-scale application is still constrained by coolant material cost, equipment compatibility and maintenance complexity [46–50].

Package-level thermal management technologies do not undertake primary heat removal. Instead, they enhance overall cooling system performance by reducing interfacial thermal resistance, strengthening heat spreading and regulating local temperature fields. Novel thermal interface materials improve heat transfer between chips and heat sinks to cut interfacial resistance, while liquid metals and similar materials suffer from leakage risks, poor interfacial stability and compatibility limitations [51,53,54]. Heat pipes, vapor chambers and phase-change materials rapidly spread heat using latent heat of phase change and effectively mitigate hotspots by homogenizing chip temperature distribution. Thermoelectric cooling is an active cooling method that further lowers hotspot temperatures at the cost of extra power consumption, and is mostly adopted as an auxiliary cooling measure. Though these package-level technologies cannot independently meet future cooling demands of ultra-high heat flux processors, they optimize heat spreading paths and reduce overall thermal resistance, serving as indispensable components of advanced liquid cooling systems.

In summary, near-junction liquid cooling focuses on boosting direct chip heat removal and acts as the core solution to address ultra-high heat flux challenges. Package-level thermal management prioritizes thermal resistance reduction, uniform temperature distribution

and device reliability, optimizing the entire heat conduction path. The two technologies are not mutually exclusive but jointly participate in processor thermal management at distinct layers. Near-junction cooling governs the speed of heat extraction from heat sources, while package-level techniques determine heat transfer efficiency inside chips and packaging assemblies. Efficient thermal management can only be realized through coordinated deployment of both approaches.

Looking ahead, advanced packaging technologies such as chiplet, 3D IC and heterogeneous integration will intensify internal thermal coupling within high-power integrated processors. Traditional design philosophies relying on standalone cooling methods will fail to meet practical demands. Chip-level thermal management research will shift from single-performance optimization to systematic collaborative optimization. Synergistic design combining high-thermal-conductivity materials, package-level heat spreaders, near-junction liquid cooling and intelligent thermal control strategies will lower chip peak temperature while balancing temperature uniformity, system reliability and energy efficiency, thereby comprehensively upgrading thermal management performance of high-power integrated processors. The multi-dimensional performance differences among various near-junction liquid cooling and packaging thermal management schemes are quantitatively summarized in Table 2, which systematically evaluates each technology from six dimensions: cooling capacity, temperature uniformity, engineering maturity, manufacturing difficulty, cost and applicable scenarios. The core merits and engineering bottlenecks of each thermal technology are further concluded in Table 3, which intuitively reflects the complementary relationship between near-junction liquid cooling and package-level thermal management. Table 4 further sorts all thermal technologies by industrial maturity level, core characteristics and application fields, forming a complete hierarchical classification system for chip thermal management.

Table 2. Comprehensive comparison of advanced chip-level thermal management technologies for high-power integrated processors.

Technology	Cooling Capability	Temperature Uniformity	Engineering Maturity	Fabrication Complexity	Relative Cost	Typical Applications
Microchannel Cooling	Very High	High	Medium	Very High	High	High-power CPUs, GPUs, AI accelerators, chiplet processors.
Jet Impingement Cooling	Very High	Medium–High	Medium	High	Medium–High	Processors with localized hotspots, HPC processors, high-power electronic devices.
Cold Plate Cooling	High	High	Very High	Medium	Medium	Data centers, servers, HPC systems, high-power processors.
Immersion Cooling	Very High	Very High	High	High	High	High-density data centers, supercomputers, large-scale computing platforms.
Thermal Interface Materials (TIMs)	Medium	Medium	Very High	Low	Low–Medium	Chip packaging, processor-to-heat-sink interfaces.
Vapor Chambers/Heat Pipes	Medium	Very High	Very High	Medium	Medium	High-performance processors, graphics cards, advanced electronic packaging.
Phase-Change Materials (PCMs)	Medium	High	Medium	Medium	Medium	Transient thermal management, portable electronics, peak heat load suppression.
Thermoelectric Cooling (TEC)	High (localized cooling)	Medium	Medium	High	High	Local hotspot suppression, precision electronic cooling, optoelectronic devices.
Multi-Physics Collaborative Thermal Management	Very High	Very High	Low–Medium	Very High	Very High	Next-generation chiplet processors, 3D ICs, heterogeneous integration, high-power integrated processors.

Table 3. Comparative summary of chip-level thermal management technologies for high-power integrated processors.

Category	Main Advantage	Main Limitation	Typical Application
Near-junction liquid cooling	Extremely high heat dissipation capability.	Complex fabrication or high system cost.	High-power processors, HPC, AI accelerators.
Package-level thermal management	Reduced thermal resistance and improved temperature uniformity.	Limited standalone cooling capacity.	Processor packaging and heterogeneous integration.

Table 4. Thermal technology classification table.

Major Thermal Technology Category	Maturity Level	Core Technical Features	Typical Application Scenarios
Cold Plate Liquid Cooling Technology	Commercial.	Embedded sealed flow channels inside metal cold plates fit closely to chip packages; circulating coolant inside dissipates heat without modifying the original chip manufacturing process.	Mainstream AI servers, general computing equipment in data centers.
Immersion Liquid Cooling Technology	Commercial (single-phase mineral oil immersion liquid cooling, etc.)/prototype (two-phase fluorinated liquid immersion liquid cooling, etc.).	Electronic components are fully immersed in dielectric cooling liquid; heat dissipation is realized via single-phase convection or boiling phase change, eliminating numerous thermal interfaces and complex external pipelines.	Supercomputing centers, high-density AI cabinets, renovation of legacy server computer rooms.
Passive Two-Phase Heat Transfer Devices (Heat Pipes/Vapor Chambers) Technology	Commercial.	Rapid heat transfer through evaporation–condensation phase-change cycles of internal working fluid; no extra pumping power consumption, simple structure and high reliability.	Medium- and low-power AI accelerator cards, edge computing boards, consumer electronic devices.
Thermal Interface Materials (TIMs) Technology	Commercial (traditional thermal grease/thermal pads, etc.)/prototype (colloid-cured liquid metal TIMs, etc.)/Emerging (gradient modified diamond composite TIMs, etc.).	Fills micron-scale gaps between chips and heat sinks to reduce contact thermal resistance; serves as a critical heat transfer medium between chips and system-level heat dissipation.	All types of AI chip packaging, thermal management supporting components for computing equipment.
Microchannel Liquid Cooling Technology	Prototype (manifold microchannel liquid cooling, etc.)/laboratory (single-phase straight fin microchannel liquid cooling, etc.).	Micron-scale flow channels fabricated on silicon or packaging substrates; coolant flows directly near heat sources to drastically shorten heat transfer paths and lower interface thermal resistance.	Prototypes of 2.5D/3D high-computing-power AI chips, custom small-size military AI chips.
Direct Jet Impingement Cooling Technology	Prototype (uniform array single jet impingement liquid cooling, etc.)/laboratory (zoned alternating inlet and outlet jet array liquid cooling, etc.).	Coolant is vertically sprayed onto chip surfaces via nozzles to rapidly break near-wall thermal boundary layers, achieving high local heat exchange and precise suppression of local hotspots.	Prototypes of medium-size single NPU computing boards, large-size multi-HBM packaged AI chip prototypes.

Table 4. *Cont.*

Major Thermal Technology Category	Maturity Level	Core Technical Features	Typical Application Scenarios
Phase-Change Materials (PCMs) Thermal Management Technology	Prototype (graphite composite organic paraffin PCMs, etc.)/laboratory (porous copper gallium-based liquid metal PCMs, etc.)/emerging (oriented thermal grid composite PCMs, etc.).	Absorbs and releases massive latent heat through solid–liquid phase change to buffer transient thermal loads and restrain drastic fluctuations of chip junction temperature.	Thermal buffering for 3D IC packaging, thermal management of computing chips with highly dynamic loads.
High-Thermal-Conductivity Material Integration Technology	Prototype (thermal vias inside silica gel sheets, etc.)/laboratory (graphene lateral heat spreading layers, etc.)/emerging (nanodiamond composite substrates, etc.).	High-thermal-conductivity components including silicon carbide, diamond and two-dimensional materials replace or reinforce traditional silicon-based structures to break the inherent thermal conductivity limit of silicon and build low-resistance high-efficiency heat transfer paths.	High-end high-power AI RF/computing chips, thermal management for 3D stacked AI chips.
3D IC/Chiplet Thermal Co-Design and Advanced Packaging Technology	Laboratory (microfluidics + TSV joint collaborative design, etc.)/emerging (full-process electrothermal coupling co-design covering architecture and packaging, etc.).	Cross-layer collaborative optimization of heat dissipation paths, packaging structures and chiplet layouts completed synchronously during chip design to resolve interlayer thermal coupling and thermal crosstalk issues of 3D stacked/chiplet architectures.	Pre-design verification for next-generation chiplet architecture AI chips, 3D stacked high-computing-power chips.

4. Chip Thermal Characterization and Temperature Measurement Technologies

4.1. Non-Contact Optical Temperature Measurement

For performance verification of chip cooling schemes, infrared thermography and micro-Raman spectroscopy represent two mainstream non-contact optical characterization methods. Obvious quantitative gaps exist between them in spatial resolution, temperature measurement depth, measurement error and time consumption. Existing research has built a complete supporting testing system based on their respective drawbacks and complementary advantages, which provides quantitative evaluation criteria for state-of-the-art heat dissipation solutions.

Infrared thermography can acquire two-dimensional full-chip temperature field data [87] and rapidly compare the overall temperature uniformity of basic cold plate and air cooling structures. Nevertheless, this technique has inherent measurement limitations. Apart from low spatial resolution that fails to identify micrometer-scale transistor hotspots, it cannot penetrate passivation layers to capture the actual junction temperature, which tends to underestimate peak temperatures near the chip junction. Micro-Raman spectroscopy compensates for the deficiencies of infrared thermography in resolution and measurement depth [88]. Bhatti and Li sorted out the applicable boundaries, errors and spatial resolution limits of infrared thermography and micro-Raman thermometry in wide-bandgap device testing, and pointed out that traditional optical testing methods still have obvious defects in ultra-wide-bandgap micro-junction measurement [89]. To break this limitation, a new thermal metrology technology for ultra-wide-bandgap semiconductor devices was developed [90], which realizes sub-micron hotspot temperature extraction

under high heat flux working conditions, making up for the insufficient detection accuracy of traditional optical characterization methods. This characterization method relies on the temperature drift effect of lattice Raman peaks to achieve ultra-high spatial resolution of 0.3–1 μm , with measurement errors controlled within $\pm 1\text{--}2\text{ }^{\circ}\text{C}$. Accordingly, a hierarchical joint testing strategy is formed: infrared thermography conducts rapid full-area hotspot localization, while micro-Raman spectroscopy performs precise fixed-point calibration. This combination effectively mitigates data distortion caused by standalone infrared temperature measurement.

Figure 34 compares temperature contours measured by infrared thermography and micro-Raman spectroscopy on the identical test area of an AlGaIn/GaN power device. The maximum hotspot temperature captured by infrared imaging is merely $114\text{ }^{\circ}\text{C}$, whereas micro-Raman characterization records a peak active-region temperature of $160\text{ }^{\circ}\text{C}$ at the same position. This contrast quantitatively demonstrates that infrared thermography significantly underestimates near-junction hotspot temperatures of electronic devices.

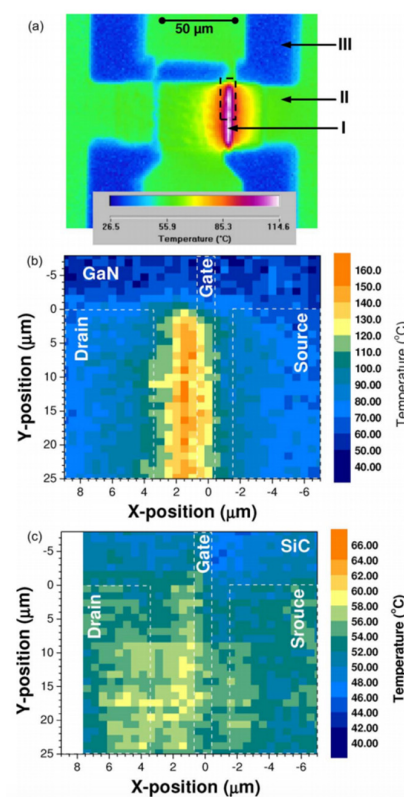


Figure 34. Comparison of temperature contours measured simultaneously by infrared thermography and micro-Raman spectroscopy for AlGaIn/GaN power devices: (a) IR temperature map of AlGaIn/GaN HFET at $V_{sd} = 40\text{ V}$ and $I = 25\text{ mA}$. Numbers denote (I) hot spot, (II) area covered by metal contacts, and (III) area free from metal contacts. (b) Raman temperature map of GaN and (c) SiC temperatures measured over the area depicted by a dashed rectangle in (a) (reproduced with permission) [88].

However, this coupled measurement method has only been verified under steady-state discrete device conditions. It fails to address the long measurement time required for single-point Raman scanning (tens of minutes) and lacks adaptability to multi-chiplet stacked architectures and liquid cooling environments. To fill the research gap regarding the applicable scope of this composite characterization system, Kim et al. [91] carried out multi-dimensional quantitative comparative experiments. They horizontally compared optical measurement methods in terms of imaging speed and detection depth, and clarified that infrared thermography is suitable for full-chip temperature screening under

dynamic operating conditions, while micro-Raman thermometry serves for high-precision calibration of static hotspots. The team also proposed a calculation method for temperature measurement error compensation under variable power loads. Nevertheless, their experiments mainly focused on conventional power devices, without measured data for cutting-edge scenarios such as AI chips and immersion cooling.

Subsequent researchers performed practical tests on commercial GPUs running deep learning workloads and validated the joint characterization method under microchannel cooling and two-phase immersion cooling conditions [92]. Experimental results reveal that standalone infrared thermography underestimates chip hotspot temperatures by 12–20 K. Optimized micro-Raman scanning trajectories can greatly reduce total testing time. The combined infrared and micro-Raman approach accurately quantifies interfacial temperature differences and thermal resistance gaps induced by various cooling channels and heat transfer fluids, establishing a standardized quantitative measurement framework tailored for high-power AI chips.

In summary, infrared thermography is suited for rapid full-chip temperature screening, whereas micro-Raman spectroscopy provides high-precision calibration of static hotspots. Their combined use enables both fast hotspot localization and accurate peak-temperature acquisition, offering a practical characterization strategy for liquid-cooled AI chips.

4.2. Embedded Contact Temperature Measurement

High-power heterogeneous chiplet processors tend to generate microscale local hotspots with spatiotemporally uneven thermal fields during operation. Non-contact measurement methods such as infrared thermography and Raman spectroscopy are obstructed by packaging layers, making it impossible to capture the actual internal temperatures of active regions, micro-bumps and interposers. Conventional wire thermocouples feature large size and high thermal capacity; their embedding introduces extra thermal resistance and disturbs the original temperature field, which prevents in situ high-precision on-chip temperature detection. Thin-film thermocouples (TFTCs) are fabricated via micro–nano thin-film deposition techniques, including magnetron sputtering and electron beam evaporation. With overall film thickness controlled within hundreds of nanometers to several micrometers, they can be integrally embedded into chip dielectric layers, interposers or active interfaces of power devices. Benefiting from low thermal disturbance, millisecond-scale dynamic response and micrometer spatial resolution, TFTCs have become the mainstream sensing solution for on-chip embedded temperature measurement [93].

The temperature measurement principle of thin-film thermocouples relies on the Seebeck thermoelectric effect. Two dissimilar thin-film electrodes form a measuring hot junction buried inside the chip and a reference cold junction led out of the package, respectively. Temperature values are retrieved by collecting the open-circuit thermoelectric potential induced by the temperature difference between the hot and cold ends. Taking the NiSi–NiCr thin-film thermocouple as an example, its structure is shown in Figure 35. This multi-layer thin-film sensor structure exhibits favorable process compatibility with standard CMOS technologies and 2.5D interposer microfabrication processes. Current on-chip integration schemes fall into three categories: conformal deposition on device surfaces, array embedding inside interposers, and insertion within packaging thermal interfaces. They support distributed multi-point temperature acquisition targeting transistor active zones, chiplet interconnect nodes and the full packaging domain, respectively [94].

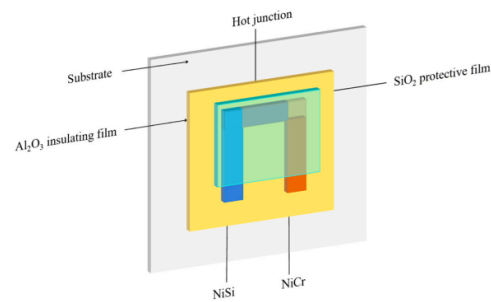


Figure 35. Schematic three-layer structure of NiCr–NiSi thin-film thermocouple (under the Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License) [94].

At the current stage, thin-film thermocouple material systems suitable for in situ chip temperature measurement are mainly divided into three categories. The Pt/Cr precious metal thin-film system features a wide linear temperature measurement range and outstanding chemical stability at high temperatures. Its Seebeck coefficient for monolithic micro-devices reaches $19.23 \mu\text{V}/^\circ\text{C}$. Embedded temperature measurement experiments on high-power GaN devices show its absolute measurement error is only $\pm 0.16^\circ\text{C}$. Nevertheless, the high cost of precious metal raw materials restricts its large-scale array deployment [95]. The NiCr/NiSi alloy thin film boasts low fabrication cost and can be deposited on polyimide flexible substrates for temperature measurement at packaging interfaces, with an effective working range from room temperature to 180°C . However, grain coarsening tends to occur under long-term high-low temperature thermal cycling, which drifts thermoelectric output signals [94]. SiGe-based semiconductor thin films are fully compatible with 300 mm standard silicon wafer processes and deliver stronger thermoelectric output. They can realize temperature sensing and tiny waste heat recovery simultaneously, yet their high intrinsic resistivity leads to a low signal-to-noise ratio, limiting their application to laboratory-scale chip thermal characterization only [96].

Embedded thin-film thermocouples possess irreplaceable technical merits. Relevant research proves that thin-film junctions have ultra-low equivalent heat capacity, so their disturbance to the chip's inherent heat dissipation paths after embedding can be neglected. Micrometer-scale arrays support simultaneous multi-point temperature sampling with a thermal response time constant below 1 ms, enabling capture of transient hotspot evolution under pulsed loads and making up for the slow dynamic response of non-contact measurement methods. Furthermore, such sensors are passive components without on-chip power supply requirements, allowing long-term online monitoring of chip thermal cycling characteristics and providing credible data support for heat dissipation performance verification of various advanced chip-level thermal management solutions [93].

Raszkowski et al.'s research team systematically studied the test error calibration and structural function identification method of embedded thermal sensors in multi-layer electronic packaging, providing standardized data processing schemes for thin-film thermocouple measurement results of high-power chiplets. Based on dual-phase-lag heat transfer theory focused on modern GAAFET structures, they first analyzed the influence of temperature and heat flux time lags on internal device temperature distribution and then established a separation algorithm for junction transient thermal resistance of stacked chips, which can separate the thermal resistance contribution of TIM layers, interposer media and TTSV vertical heat conduction paths from the overall test curve, realizing quantitative decomposition of multi-series thermal interfaces in advanced packaging [97]. The team also verified the thermal interference of thin-film sensor arrays on the original chip temperature field through steady-state and transient comparison experiments, and proposed a low-disturbance embedded layout optimization rule for distributed temperature measurement

arrays, which effectively suppresses measurement deviation caused by sensor-induced extra thermal capacitance in 3D IC multi-stack structures.

Nevertheless, this technology still suffers from several technical drawbacks. The sealed packaging cavity fails to provide a stable cold-junction reference, and the absence of cold-junction compensation introduces systematic measurement errors. High-density wiring easily triggers signal crosstalk, and multi-channel synchronous readout slows down temperature response. The mismatch of thermal expansion coefficients between thin-film electrodes and silicon substrates causes film delamination and interfacial cracking under thermal cycling ranging from $-40\text{ }^{\circ}\text{C}$ to $125\text{ }^{\circ}\text{C}$, resulting in permanent degradation of thermoelectric performance. Fabricating thin-film arrays after front-end chip processes may damage transistors during lithography and sputtering, raising the packaging cost and reducing the yield of 2.5D chiplet devices [96].

In recent years, domestic and foreign scholars have proposed a series of optimization strategies to address the above limitations. Guo Ke and other researchers [98] introduced a NiCrAlY transition layer to match the thermal expansion coefficient of the substrate, adopted dual processes to fabricate an alumina insulating protective film, and optimized the thickness and sputtering parameters of the Pt–Rh sensitive layer. These measures alleviate film delamination, cracking and large measurement errors. Meanwhile, thin films are directly deposited on nickel-alloy substrates to reduce thermal inertia and accelerate response, improving sensor stability and measurement accuracy at high temperatures.

From the perspective of technological development trends, embedded thin-film thermocouples will serve as the core sensing component of closed-loop characterization systems for thermal management in high-power integrated processors. High-density thin-film arrays can quantitatively evaluate the temperature control performance of advanced on-chip cooling technologies such as microchannel liquid cooling and on-chip thermoelectric refrigeration. Long-term in situ temperature measurement data can support the construction of digital twin models for chip thermal failure. Future research will focus on three directions: CMOS-compatible low-cost thermoelectric film systems, low-crosstalk high-density multi-channel readout circuits, and composite film structures with high thermal cycling resistance, so as to further expand the engineering application scope of in situ temperature measurement for large-computing-power AI chiplet chips.

Accurate quantification of microscale hotspots in wide-bandgap and AI chips remains a critical challenge in thermal characterization. Existing studies have proposed a junction temperature extraction method for GaN devices based on turn-off delay. This method enables acquisition of actual temperatures within micrometer-scale active regions without damaging packaging structures, addressing the limitation that infrared thermometry underestimates hotspot temperatures. Furthermore, it can be integrated with embedded thin-film thermocouples to establish a combined steady-state and transient measurement system [99].

4.3. Synergistic Verification of Thermal Testing and Simulation Models

The development of chip-level cooling technologies relies not only on innovative heat-sink structures and advanced materials, but also on accurate thermal characterization and predictive capabilities. Whether evaluating multi-layer interfacial resistance in packaged devices, validating the cooling limits of microchannels or jet impingement schemes, or assessing the transient response of wide-bandgap power components, experimental measurements and numerical simulations must be jointly employed to establish a closed-loop “test–model–validate” workflow. Thermal testing methodologies and simulation tools therefore constitute indispensable components of the overall thermal management toolkit for high-power integrated processors.

The importance of interfacial thermal resistance characterization has been systematically addressed in the context of electronic packaging, where precise quantification of the chip-to-heat-sink contact resistance is a prerequisite for assessing thermal interface material performance and package efficiency [100]. A key advance highlighted in that work is the demonstration that thermal transient testing offers clear advantages over steady-state methods, particularly for extracting interface properties with higher sensitivity. Building on this foundation, the same measurement paradigm has been extended to wide-bandgap devices, where improved transient test protocols have been developed to cope with the extreme heat-flux conditions typical of SiC and GaN components [101]. These efforts collectively show that transient thermometry, when properly adapted to actual device operating modes, provides a reliable experimental basis for quantifying internal thermal paths and interface bottlenecks, and for verifying the actual temperature reduction achieved by various cooling interventions.

On the modeling side, the need for fast yet accurate thermal simulation at the chip layout level has led to dedicated tools that enable rapid temperature-field computation during early design stages, thereby facilitating thermal-aware floor planning without heavy computational overhead [102]. For more complex scenarios, such as ultra-thin stacked chips with multiple time-varying heat sources, conventional numerical methods often become prohibitively expensive in the time domain. This challenge has been tackled by introducing fast non-linear dynamic compact models that preserve fidelity while drastically reducing simulation runtime, thus making transient thermal analysis of 3D IC and chiplet architectures practically feasible [103].

When viewed together, the experimental methods and simulation frameworks described above are not isolated alternatives but complementary pillars. High-precision transient testing (exemplified by the work on interfacial resistance and wide-bandgap devices) supplies benchmark data for model calibration, whereas advanced simulation tools (developed for both steady-state layout inspection and dynamic multi-source problems) extend the spatial and temporal resolution beyond what measurements alone can offer. It is this synergy—where experiments anchor the models and simulations guide the measurement strategy—that constitutes the methodological backbone for reliable thermal design and performance validation in high-power heterogeneous processors.

5. Conclusions

This paper takes chip-level thermal management technologies for high-power integrated processors as the research object, systematically reviews the mechanisms, development trajectories, and application characteristics of current mainstream heat dissipation technologies, summarizes the current technical bottlenecks in light of the thermal issues caused by increasing power consumption and architectural evolution of high-power integrated processors, and identifies future development trends in the field. The main conclusions are as follows:

1. The thermal environment of high-power integrated processors is becoming increasingly complex, and traditional cooling methods can no longer meet application requirements.

Driven by large model computing upgrades, high-power integrated processors see rising power consumption and local heat flux, with peak values up to 1 kW/cm². Widely adopted 2.5D/3D chiplet heterogeneous integration and 3D IC stacking further aggravate interlayer thermal resistance, thermal coupling and cross-die thermal crosstalk, forming complex multi-scale and non-uniform thermal fields. Constrained by low heat transfer capability, traditional air cooling features insufficient cooling capacity, high energy consumption and unstable temperature regulation, and can no longer support high-power

integrated processor devices. Accordingly, liquid cooling has become an indispensable solution for ultra-high heat flux dissipation.

2. The near-junction liquid cooling technology system is gradually maturing, with different technical routes offering differentiated suitability for various application scenarios.

Microchannel liquid cooling has evolved through structural iterations covering single-phase heat transfer, phase-change heat transfer, and manifold distribution modes. Among them, three-dimensional manifold microchannels well adapt to chiplet and 3D stacking architectures, serving as a core technique for ultra-high heat flux cooling. Jet impingement cooling enables localized heat transfer enhancement and precise hotspot suppression, while segmented alternating inlet/outlet structures effectively alleviate crossflow interference. As a mature and highly engineering-adaptable scheme, cold plate liquid cooling achieves improved temperature uniformity via grid manifold and topological optimization, and has become the dominant cooling approach for data centers. In addition, immersion liquid cooling realizes system-wide uniform temperature control and reduces data center PUE. Combined with heat pipes, passive immersion cooling supports pumpless heat dissipation for kilowatt-level integrated processors, showing great applicability in high-density supercomputing clusters.

3. Package-level thermal management technologies have formed a diversified technology matrix, providing comprehensive support for reliable integrated processor operation.

Thermal interface materials have evolved from conventional polymers to liquid metals, diamond composites and structural composite materials, effectively mitigating interfacial thermal resistance, medium leakage and packaging compatibility issues. Heat pipes and vapor chambers realize efficient heat transfer and two-dimensional temperature uniformity via phase-change heat transfer, with miniaturization and ultra-thinning as key development trends. Phase-change materials (PCMs) suppress transient temperature fluctuations through latent heat storage, and innovative composite PCMs overcome the trade-off between high thermal storage capacity and high thermal conductivity. High-thermal-conductivity substrates such as silicon carbide and diamond, integrated with embedded thermal structures including thermal through-silicon vias and graphene, break the thermal conductivity limitations of traditional silicon-based materials through material and structural optimization. For wide-bandgap and ultra-wide-bandgap power devices, heterogeneous substrate integration remains the most effective near-junction heat dissipation route, and the related device-level thermal management technical system has been comprehensively reviewed by Zhou et al. [104], providing a reference for the thermal design of high-power compound semiconductor chips. For 3D IC and chiplet advanced packaging of high-power integrated processors, thermal co-design methods integrating electrothermal coupling analysis and architecture-packaging co-optimization have become the mainstream design paradigm.

4. Existing thermal management technologies still face numerous technical bottlenecks in engineering implementation.

Frontier near-junction liquid cooling technologies such as microchannels and jet impingement suffer from complex microfabrication processes, high manufacturing costs, and insufficient compatibility with advanced packaging processes. Various liquid cooling systems generally face risks such as coolant leakage, channel corrosion, and insufficient long-term operational reliability. In the packaging domain, high-thermal-conductivity composite materials and thermal interface structures remain constrained by factors such as interfacial thermal resistance, thermal expansion coefficient mismatch, and long-term aging degradation. Furthermore, a single heat dissipation technology is often inadequate to

match the non-uniform heat generation characteristics of high-power integrated processors, and the cross-level co-design capability integrating processor, package, and system levels still needs improvement.

5. The industry technology development direction is clear, with high integration, high reliability, and co-design becoming the core evolutionary goals.

In light of the development trends of high-power integrated processors, 3D stacking, and chiplet heterogeneous integration, chip-level thermal management technologies will continue to evolve toward the goals of “high integration, low thermal resistance, low flow resistance, and high operational reliability”. Future research needs to focus on key technical breakthroughs in areas such as topological optimization of microchannel structures, low-leakage hermetic packaging processes, and novel high-thermal-conductivity composite materials, while advancing cross-level thermal co-design through multi-physics coupling at the processor, package, and system levels, thereby promoting the transition of various advanced thermal management technologies from laboratory-scale fundamental research to large-scale engineering applications.

6. Challenges and Outlook

Focusing on the development needs of near-junction liquid cooling and package-level thermal management for high-power integrated processors, this paper systematically reviews mainstream technological approaches including microchannel liquid cooling, direct jet impingement cooling, cold plate liquid cooling, and immersion liquid cooling, as well as package-level thermal interface materials, heat pipes/vapor chambers, and high-thermal-conductivity substrates. The working mechanisms, structural characteristics, and research progress of various heat dissipation technologies are elucidated, providing theoretical references and technical support for the selection and structural design of thermal management solutions for high-power integrated processors. Although various near-junction liquid cooling and package-level heat dissipation technologies currently exhibit excellent high-heat-flux adaptability, numerous practical challenges remain in their engineering implementation and large-scale application. On the one hand, near-junction liquid cooling structures such as microchannels and jet impingement remain largely at the laboratory research stage, facing issues including complex microfabrication processes, high manufacturing costs, and significant difficulties in achieving compatibility with chiplet and 3D IC advanced packaging technologies. On the other hand, liquid cooling systems generally suffer from risks such as fluid leakage, channel corrosion, and insufficient long-term operational reliability, while package-level high-thermal-conductivity materials and thermal interface structures remain constrained by factors such as interfacial thermal resistance, thermal mismatch, and aging degradation.

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